



**DESIGN AND IMPLEMENTATION OF A MULTIMODAL OFFLINE VOICE-
CONTROLLED PATIENT ASSISTANCE SYSTEM WITH GSM EMERGENCY
ALERTING**

BY

ABUBAKAR ZUBAIR OKHAYOLE

ENG/19/ELE/00265

SUPERVISED BY

PROF. BALA BOYI BUKATA

MAY, 2026

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A PROJECT SUBMITTED TO THE DEPARTMENT OF ELECTRICAL ENGINEERING,
BAYERO UNIVERSITY, KANO, IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR THE AWARD OF BACHELOR DEGREE IN ELECTRICAL
ENGINEERING (B.ENG. ELECTRICAL)

SUPERVISED BY

PROF. BALA BOYI BUKATA

MAY, 2026

DECLARATION

I hereby declare that this project report is the product of my own research efforts except for quotation and citations which have been duly acknowledged, undertaken under the supervision of Prof. Bala Boyi Bukata and has not been presented and will not be presented elsewhere for the award of a degree or certificate.

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CERTIFICATION

I hereby certify that I have read this project report by ABUBAKAR ZUBAIR OKHAYOLE (ENG/19/ELE/00265) and it was carried out under my supervision and in my opinion this project is sufficient in terms of scope and quality for the award of the degree of Bachelor of Engineering (Electrical).

Signature:

Name: Prof. Bala Boyi Bukata

Date:

APPROVAL

This is to certify that this research project is an original work undertaken by ABUBAKAR ZUBAIR OKHAYOLE (ENG/19/ELE/00265) and has been prepared in accordance with the regulation governing the preparation and presentation of undergraduate project in Bayero University, Kano.

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To my family: your prayers, encouragement, and sacrifices are the bedrock upon which everything I achieve is built. Words cannot adequately express my gratitude.

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Finally, I acknowledge the open-source community (the developers of the Arduino framework, Flutter, the ESP32 ecosystem, and the numerous libraries upon which this work builds) whose contributions make advanced embedded systems accessible to researchers worldwide.

DEDICATION

To my family, whose unwavering support and prayers sustained me through every stage of this journey, my father who we lost when it was very close to almost done, my mother who has never given up on any of us for a second, and to the patients in resource-constrained hospitals whose dignity and safety inspired this work.

ABSTRACT

Mobility-impaired patients in Nigerian hospitals face a critical patient safety gap: conventional nurse call systems require physical button activation inaccessible to patients with limited motor function, while commercially available voice-controlled alternatives are uniformly dependent on stable internet connectivity unavailable in many Nigerian healthcare facilities. This research addresses this gap through the design and implementation of a Multimodal Offline Voice-Controlled Patient Assistance System with GSM Emergency Alerting, specifically engineered for deployment in resource-constrained healthcare environments. The system integrates an Ai-Thinker VC-02 Edge Artificial Intelligence module for fully offline speech recognition, an ESP32 microcontroller implementing a centralised state machine firmware architecture, a SIM800L GSM module for SMS-based emergency alerting independent of hospital network infrastructure, relay-switched mains-voltage appliance control via an LTV-247 optocoupler and BC237 transistor driver circuit, and a Flutter-based companion mobile application providing Bluetooth Low Energy redundant control with real-time state synchronisation. Physical pushbutton switches provide a third independent control modality, ensuring system operability in the complete absence of electronic intelligence. A culturally adaptive multi-phrase command vocabulary accommodates the natural English speech patterns of Nigerian patients across seven command categories, with wake-word-free emergency command detection enabling immediate alert triggering without prior activation sequences. The modular firmware architecture, designated v4.0, incorporates non-blocking subsystem implementations, persistent NVS-based configuration storage, BLE ACK/NACK communication protocol, and a 25-entry ring-buffer event logging system. Experimental evaluation demonstrated voice recognition accuracy of 100% under both quiet and moderate ambient noise conditions, emergency response latency of approximately 0.6 seconds mean, 100% SMS delivery success rate on the Airtel Nigeria network, 100% BLE command delivery reliability, and 100% state synchronisation accuracy across all modality combinations. Battery life of approximately 5.68 hours was estimated under continuous idle operation from a 2000mAh 18650 lithium-ion cell with MT3608 boost conversion. Comparative analysis confirms that the present system exceeds the performance of comparable offline systems in the reviewed literature while uniquely combining offline operation, GSM emergency alerting, and multimodal control within a single integrated patient assistance platform. The system demonstrates that clinically relevant assistive capabilities can be delivered reliably in resource-constrained environments without dependence on internet infrastructure.

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LIST OF ACRONYMS

<i>Acronym</i>	<i>Definition</i>
AC	Alternating Current
ACK	Acknowledgement
ADC	Analogue-to-Digital Converter
AI	Artificial Intelligence
APK	Android Package Kit
AT	Attention (Hayes Command Set)
BLE	Bluetooth Low Energy
BOD	Brownout Detection
CC-CV	Constant-Current Constant-Voltage
CTR	Current Transfer Ratio
DC	Direct Current
Edge AI	Edge Artificial Intelligence
ESP32	Espressif Systems IoT Microcontroller Module 32
GATT	Generic Attribute Profile
GPIO	General Purpose Input/Output
GPRS	General Packet Radio Service
GSM	Global System for Mobile Communications
IDE	Integrated Development Environment
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
LED	Light Emitting Diode
NACK	Negative Acknowledgement
NVS	Non-Volatile Storage
PCB	Printed Circuit Board
PWM	Pulse Width Modulation
SMS	Short Message Service
UART	Universal Asynchronous Receiver-Transmitter
UUID	Universally Unique Identifier
VC-02	Voice Control Module 02 (Ai-Thinker)
WiFi	Wireless Fidelity

LIST OF SYMBOLS

Symbol	Definition
V _{out}	MT3608 boost converter output voltage (V)
V _{ref}	MT3608 internal feedback reference voltage (V)
V _{in}	MT3608 input voltage / battery terminal voltage (V)
D	Duty cycle of the MT3608 boost converter
I _{coil}	Relay coil energisation current (A)
R _{coil}	Relay coil resistance (Ohm)
I _C	Transistor collector current (A)
I _B	Transistor base drive current (A)
h _{FE}	Transistor DC current gain (dimensionless)
V _{BE}	Transistor base-emitter voltage (V)
V _{CE(sat)}	Transistor collector-emitter saturation voltage (V)
SF	Transistor saturation factor (dimensionless)
I _F	Optocoupler LED forward current (mA)
V _F	Optocoupler LED forward voltage (V)
CTR	Optocoupler current transfer ratio (%)
I _{pullup}	GPIO pull-up resistor quiescent current (mA)
Q	Charge required during SIM800L transmission burst (C)
DeltaV	Allowable terminal voltage droop (V)
C _{min}	Minimum decoupling capacitance required (F)
t	Estimated battery operating duration (h)
eta	MT3608 boost converter efficiency (%)
I _{fan}	AC fan load current (A)
I _{bulb}	AC bulb load current (A)
P	Appliance power rating (W)
V _{AC}	Nigerian mains supply voltage (V)

CHAPTER ONE

INTRODUCTION

1.1 Background of Study

The delivery of safe and dignified patient care within hospital environments depends critically on the ability of patients to communicate their needs to medical personnel promptly and reliably. For patients experiencing mobility impairment (whether resulting from surgical procedures, acute illness, neurological conditions, or traumatic injury), this communication is conventionally mediated through nurse call systems that require the patient to physically locate and press a designated call button [1]. When a patient is incapacitated, restrained, or experiencing acute physical distress, such systems become practically inaccessible, creating a dangerous gap between patient need and clinical response.

The World Health Organization estimates that over one billion people globally live with some form of disability, with mobility-related impairments representing one of the most prevalent categories across all age groups [2]. Within hospital inpatient populations specifically, temporary mobility limitation is considerably more common than in the general population, affecting patients recovering from orthopaedic surgeries, neurological events, and a broad range of acute medical conditions. Despite this prevalence, the fundamental interaction paradigm of nurse call technology has remained largely unchanged for decades, requiring physical button activation that presupposes a degree of motor function many patients do not possess [3].

Voice-activated assistive technologies have emerged globally as a promising solution to this challenge. The rapid proliferation of consumer voice assistants (including Amazon Alexa, Google Assistant, and Apple Siri) has demonstrated the practical viability of natural language interfaces for environmental control, information retrieval, and communication tasks. Subsequent research has explored the application of these platforms within healthcare settings, with studies reporting measurable reductions in patient distress and improvements in care efficiency where voice-controlled systems have been deployed [4]. However, these commercial platforms share a fundamental architectural dependency: all speech processing is performed on remote cloud servers, requiring continuous and stable internet connectivity to function.

This dependency represents a critical barrier to deployment in a substantial proportion of global healthcare facilities. Nigeria, with a population exceeding 220 million and a public healthcare system serving the majority of this population, exemplifies the infrastructure constraints characteristic of many developing healthcare systems. Oluwale et al. [5] documented the pervasive impact of inadequate technological infrastructure on healthcare quality in Nigeria, identifying unreliable power supply and internet connectivity as primary constraints on the adoption of digital health technologies. Umar et al. [6] further identified the absence of reliable communication infrastructure as a systemic barrier to technology-assisted healthcare delivery in Nigerian primary and secondary care facilities. The Nigerian Communications Commission reported a national internet penetration rate of approximately 45.57% as of 2023, with connectivity quality varying substantially across geographic regions and facility types [7].

Edge Artificial Intelligence represents a technically mature and practically viable alternative to cloud-dependent speech processing. By executing machine learning inference directly on embedded hardware at the point of use, Edge AI systems eliminate the dependency on external connectivity, reduce response latency to milliseconds, and ensure the complete privacy of processed data. Rancea et al. [8] comprehensively reviewed Edge AI deployments in healthcare contexts, identifying offline speech recognition as among the most promising applications. Abdellatif et al. [9] similarly identified Edge AI as a transformative enabler for smart healthcare in resource-constrained environments. The system developed and evaluated in this research (the Multimodal Offline Voice-Controlled Patient Assistance System with GSM Emergency Alerting) addresses the convergence of these challenges by integrating Edge AI-based offline speech recognition with GSM cellular communication, relay-based mains-voltage appliance control, Bluetooth Low Energy mobile application connectivity, and physical switch redundancy within a unified state-managed architecture.

1.2 Statement of Problem

Despite significant global advances in assistive technology and voice-controlled systems, mobility-impaired patients in Nigerian and other resource-constrained healthcare facilities remain substantially underserved by existing nurse call and environmental control technologies. The following specific problems motivate this research:

First, conventional nurse call systems universally require physical button activation, rendering them inaccessible to patients whose medical conditions preclude deliberate physical interaction

[3]. Raut et al. [1] identified this physical interaction requirement as the primary accessibility limitation of existing patient assistance infrastructure, noting that the patients most urgently requiring assistance are precisely those least able to activate conventional systems.

Second, commercially available voice-controlled assistive platforms require stable internet connectivity for cloud-based speech processing. Chithra et al. [10] explicitly identified cloud dependency as the principal limitation of mainstream voice assistant deployments in healthcare. In Nigerian healthcare facilities specifically, where internet connectivity is documented as unreliable and frequently unavailable [5][6], cloud-dependent voice systems cannot be considered viable patient safety infrastructure.

Third, existing voice-controlled systems employ rigid single-phrase command recognition, requiring users to memorise exact trigger phrases. Kumar [11] highlighted the cognitive burden imposed by inflexible command vocabularies on elderly and distressed users, demonstrating a significant correlation between command naturalness and successful system adoption.

Fourth, the absence of reliable emergency alerting mechanisms that function independently of hospital network infrastructure represents a critical patient safety gap. Al-khafajiy et al. [12] documented that conventional digital nurse call systems dependent on hospital network infrastructure exhibited complete failure during network outages, precisely the scenarios in which reliable emergency alerting is most essential. GSM cellular networks, by contrast, maintain independent communication infrastructure that continues to function during local power and network failures.

Fifth, existing assistive systems rarely combine environmental control and emergency alerting within a cohesive, redundant architecture. The absence of multi-modal redundancy (wherein voice, mobile application, and physical switch control modalities operate synchronously with unified state management) means that single points of failure in any individual control pathway can render the entire assistance system inoperable [13].

1.3 Aim and Objectives

The aim of this research is to design and implement a multimodal offline voice-controlled patient assistance system with GSM emergency alerting.

The following specific objectives are pursued to achieve this aim:

1. To design and implement an offline Edge AI voice recognition subsystem capable of recognising multiple English-language command variations reflecting the natural speech patterns of Nigerian patients, without internet connectivity.
2. To design and implement a wake-word-free emergency response subsystem that detects distress voice commands immediately and triggers alerts to medical personnel via GSM-based SMS notification and local buzzer alarm.
3. To design and implement a relay-based mains-voltage appliance control unit for hospital room environmental devices (specifically lighting and ventilation) with provision for future expansion to additional electrical loads.
4. To develop and integrate a Bluetooth Low Energy companion mobile application providing redundant access and real-time state monitoring, ensuring compatibility with voice and manual control modalities.
5. To evaluate system performance with respect to voice recognition accuracy, emergency response latency, GSM alert reliability, BLE communication stability, and overall operational consistency.

1.4 Significance of Study

This research makes contributions of significance across clinical, technological, and engineering dimensions.

From a clinical perspective, the system directly addresses a documented patient safety gap by enabling mobility-impaired individuals to independently summon medical assistance and control their immediate environment without physical intervention. Morris et al. [14] demonstrated that patient autonomy in environmental control is significantly correlated with reduced anxiety, improved sleep quality, and greater overall satisfaction with care, outcomes with direct implications for clinical recovery trajectories. The system's wake-word-free emergency detection capability eliminates the cognitive and temporal overhead of wake word activation in scenarios where every second of response latency may be clinically consequential.

From a technological perspective, the research demonstrates the practical feasibility of deploying sophisticated AI-powered assistive systems in infrastructure-limited environments through the strategic application of Edge AI, establishing a replicable design pattern for offline intelligent systems in resource-constrained contexts. From an engineering perspective, the

multi-modal redundancy architecture (integrating voice recognition, mobile application control, relay-switched mains-voltage load control, and physical switching with centralised state management) contributes a robust design pattern for safety-critical embedded systems. Xia et al. [4] identified multi-modal redundancy as a key determinant of patient assistance system reliability in underserved healthcare settings.

From a socioeconomic perspective, the system's design prioritises affordability and local relevance, employing widely available, low-cost embedded hardware components and the Airtel GSM network for emergency communication. This economic accessibility is essential for meaningful adoption within the Nigerian public healthcare system, where technology procurement budgets are severely constrained [5][6].

1.5 Scope of Study

This research encompasses the design, implementation, and experimental evaluation of a functional prototype patient assistance system. The scope is defined as follows:

The system is designed for single-bed deployment, controlling two mains-voltage appliances (a 6W room light and a 60W ventilation fan connected to the 240V AC Nigerian mains supply) through two channels of a relay module driven by an LTV-247 optocoupler and BC237 transistor relay driver circuit. The remaining two relay channels are reserved for future expansion without hardware modification.

Voice recognition is implemented using the Ai-Thinker VC-02 Edge AI module [15], supporting multiple English language phrase variations across seven primary command categories. Each command category accommodates between two and five recognised phrase variants to accommodate natural speech diversity. Support for non-English Nigerian languages is identified as future work.

Emergency and nurse-call alerting is implemented via SMS transmission over the Airtel GSM network, utilising the SIM800L GSM module [16]. The companion mobile application is developed using the Flutter framework [17] for the Android platform. System evaluation is conducted under controlled conditions simulating a hospital room environment. Clinical trials involving actual patients are beyond the scope of this prototype study and would require independent institutional ethics board approval.

1.6 Outline of Report

This report is organised into five chapters as follows: Chapter One provides the background context, problem statement, research aim and objectives, significance of the study, scope of work, and structural outline of the report. Chapter Two presents a comprehensive review of related literature encompassing assistive technology in healthcare, voice-controlled systems in medical environments, Edge AI and offline speech recognition, GSM-based emergency alerting, IoT applications in resource-constrained healthcare, mobile application development for patient care, and an analysis of the research gap addressed by this work. Chapter Three details the system design and methodology, covering the overall system architecture and block diagram, hardware component selection and justification over alternatives, complete circuit design, firmware architecture, voice command configuration, mobile application design, and eleven design calculations validating each major circuit parameter. Chapter Four presents the implementation process and experimental evaluation results across voice recognition accuracy, emergency response latency, SMS delivery reliability, Bluetooth communication stability, and power consumption. Chapter Five concludes the report with a synthesis of research findings, identification of system limitations, recommendations for improvement, and directions for future research.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a structured review of existing literature relevant to the design and implementation of the proposed patient assistance system. The review is organised across ten thematic areas: assistive technology in healthcare environments, voice-controlled systems in medical settings, Edge Artificial Intelligence and offline speech recognition, GSM-based emergency alerting systems, nurse call system evolution and limitations, Bluetooth Low Energy in healthcare applications, Internet of Things and home automation foundational approaches, mobile applications in patient care, multimodal interaction design for elderly and mobility-impaired users, and healthcare technology adoption in Nigeria. Each section synthesises findings from peer-reviewed literature and established technical documentation, culminating in a clear articulation of the research gap that the present work addresses.

2.2 Assistive Technology in Healthcare Environments

Assistive technology is broadly defined as any device, software, or system that maintains or improves the functional capabilities of individuals with physical, sensory, or cognitive impairments [1]. Within hospital environments, assistive technologies serve a dual clinical purpose: preserving patient autonomy and dignity by enabling self-directed interaction with the care environment, and reducing the volume of routine nursing interventions required to meet patient needs, thereby allowing clinical staff to allocate their time more effectively toward complex care tasks [14].

The World Health Organization recognises assistive technology as a fundamental component of universal health coverage, estimating that over one billion people globally require at least one assistive product, with this figure projected to exceed two billion by 2050 as populations age and the prevalence of non-communicable diseases increases [2]. Despite this recognised need, the WHO further estimates that only one in ten people who require assistive technology currently have access to it, a disparity particularly acute in low- and middle-income countries where healthcare infrastructure constraints limit both the procurement and maintenance of assistive devices.

Raut et al. [1] conducted a comprehensive review of assistive technology applications for mobility-impaired individuals, documenting that the consequences of inadequate assistive provision extend beyond physical inconvenience to encompass measurable psychological impacts including increased anxiety, reduced sense of agency, and diminished treatment adherence. The authors specifically identified nurse call systems as the most critical assistive technology in acute care environments, noting that delays in nurse response attributable to call system inaccessibility were associated with adverse patient outcomes including falls, pressure injuries, and deteriorating clinical status.

Morris et al. [14] investigated user perspectives on assistive device design through structured interviews with mobility-impaired individuals across multiple care settings, identifying reliability, ease of activation, and minimal cognitive burden as the three most consistently prioritised design attributes. Participants expressed strong preferences for systems that accommodated natural interaction modalities (particularly voice) over those requiring deliberate physical manipulation, reinforcing the clinical rationale for voice-controlled assistance in patient populations with varying degrees of motor function.

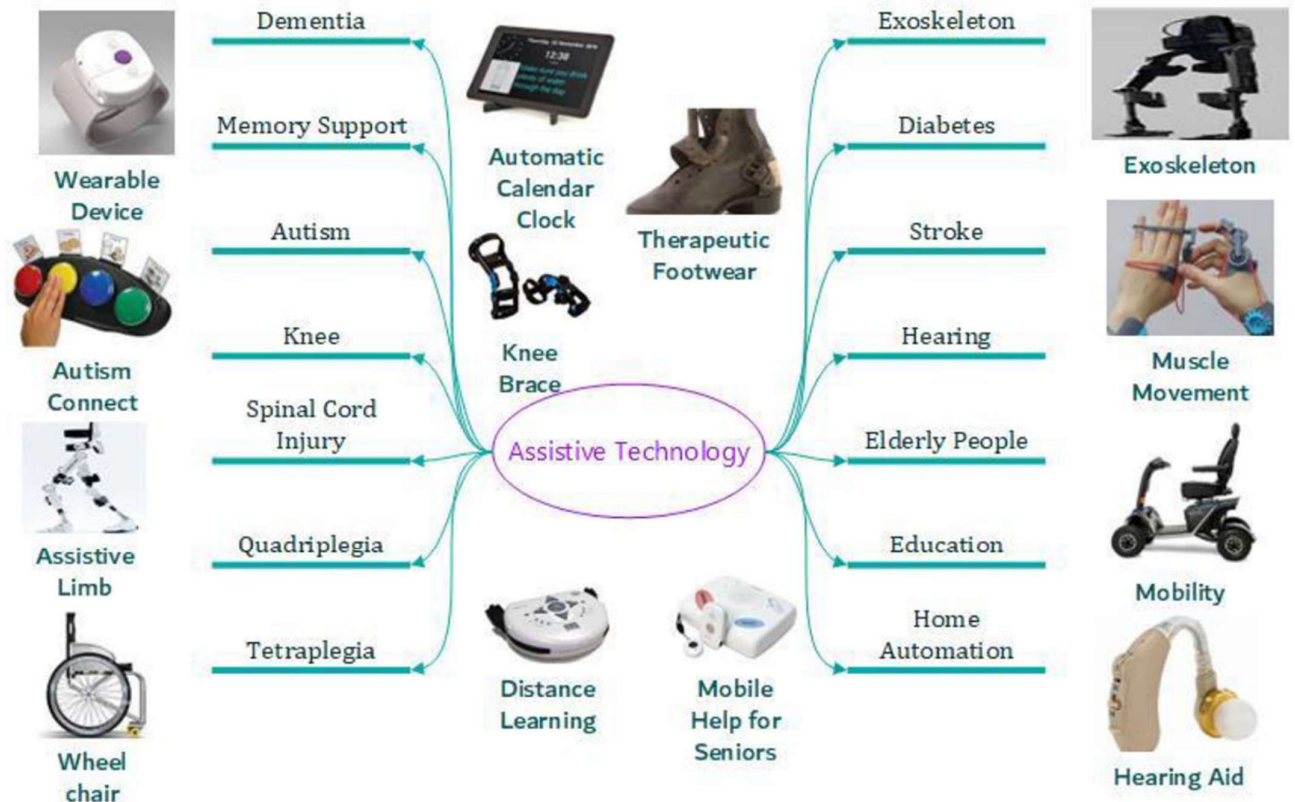


Figure 2.1: Assistive Technology Application Fields

Goher and Fadlallah [18] reviewed assistive devices specifically designed for elderly mobility and rehabilitation, noting that the ageing global population was driving significant demand for technologies that could support independent function in both community and institutional care settings. The authors identified voice control as an emerging priority in assistive device design for elderly users, citing its independence from fine motor skills that frequently decline with age as a key advantage over touch-based interaction modalities. The nurse call system has undergone considerable technical evolution since its inception, progressing from simple wired bell systems to sophisticated digital platforms capable of routing calls to specific personnel, logging response metrics, and integrating with electronic health record systems [3]. However, Sharma and Gautam [3] noted that this technological sophistication had been directed primarily toward back-end infrastructure rather than toward the patient-facing activation interface, which has remained fundamentally unchanged in its requirement for deliberate physical button activation.

2.3 Voice-Controlled Systems in Medical Environments

The application of voice control technology to patient-facing healthcare systems has been an active area of research and development for over a decade, driven by the dual imperatives of improving accessibility for patients with limited motor function and reducing the cognitive burden associated with manual interaction with care environment controls [11].

Kumar [11] conducted a comprehensive analysis of speech recognition systems in healthcare, reviewing developments across clinical documentation, patient monitoring, and patient assistance domains. The study identified three principal limitations constraining the adoption of voice-controlled systems in healthcare environments: first, the dependency of high-performance systems on cloud-based processing infrastructure; second, the inflexibility of command vocabularies that require users to reproduce exact trigger phrases; and third, the absence of robust emergency detection mechanisms capable of triggering responses without the cognitive overhead of wake word activation sequences. These limitations directly informed the design objectives of the present research.

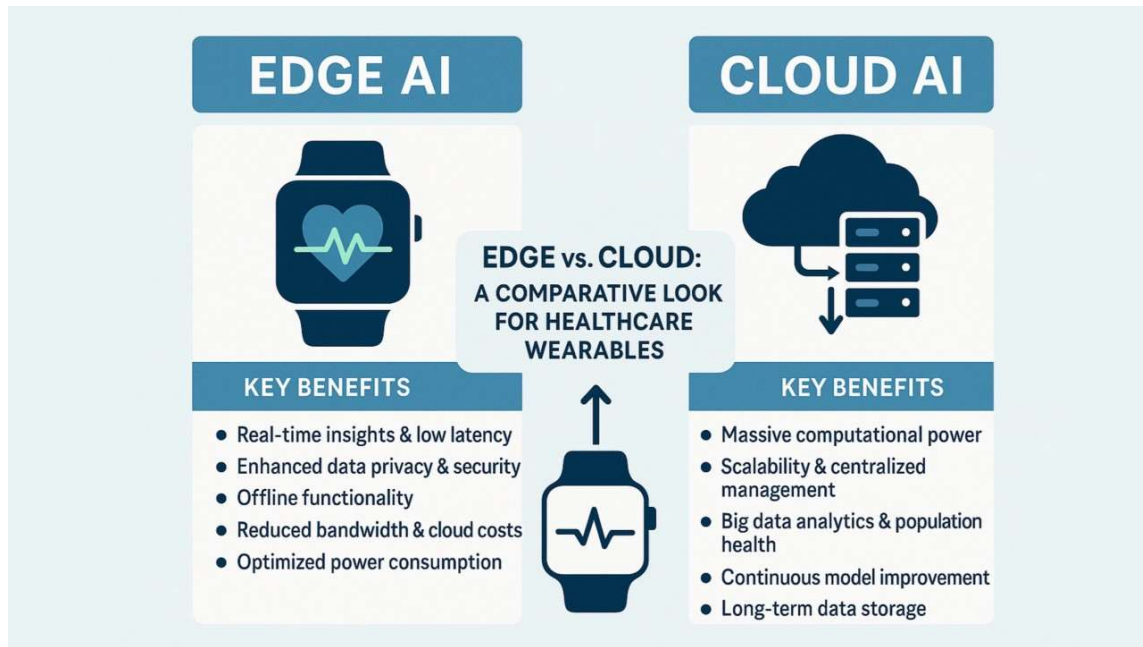


Figure 2.2: Cloud-Based vs. Edge AI Architecture Comparison

Chithra et al. [10] developed a voice health assistant specifically targeting elderly users, incorporating offline processing capability and multi-language support as primary design objectives. The system demonstrated that offline speech recognition could achieve clinically acceptable accuracy levels without cloud connectivity, representing a significant advancement over prior cloud-dependent implementations. However, their implementation did not incorporate emergency alerting functionality or integration with nursing communication infrastructure, limiting its applicability as a comprehensive patient assistance platform.

Kar et al. [19] proposed an intelligent voice automation system with dual online and offline operational modes. The authors acknowledged that the offline mode supported a significantly reduced command vocabulary, potentially excluding emergency-related commands precisely when reliable offline operation was most critical, a limitation that the present research addresses by implementing full command vocabulary support entirely offline. Kiran et al. [20] developed an offline virtual voice assistant with system automation capabilities, reporting recognition accuracy of 93.7% across a forty-entry command vocabulary. However, the system was designed for general consumer use and lacked the emergency alerting, GSM integration, and clinical environmental control capabilities required for patient assistance applications. Xia et al. [4] specifically examined voice-based healthcare solutions for patients in underserved areas, calling for the development of infrastructure-independent voice healthcare solutions as a priority research direction, a call directly addressed by the present work. Pukdesree and Pukdesree [21] demonstrated the technical viability of voice command to physical appliance

actuation within an embedded IoT architecture, though their system focused on domestic environments and omitted emergency alerting functionality.

2.4 Edge Artificial Intelligence and Offline Speech Recognition

Edge Artificial Intelligence refers to the execution of machine learning inference directly on end-user devices or local embedded hardware, as opposed to transmitting data to centralised cloud servers for remote processing [8]. This architectural paradigm has emerged as a critical enabler for intelligent applications in environments where reliable connectivity cannot be assumed, and where the latency of round-trip cloud communication is incompatible with real-time responsiveness requirements [9].

Rancea et al. [8] provided a comprehensive review of Edge AI deployments in healthcare, documenting three primary advantages: operational independence from network infrastructure, elimination of data transmission latency enabling real-time system responsiveness, and enhanced patient data privacy through localised processing. Abdellatif et al. [9] examined Edge AI within the context of smart health systems in resource-constrained environments, identifying offline keyword recognition as among the highest-priority Edge AI capabilities for healthcare IoT deployment, noting that emergency command detection (requiring sub-second response times and absolute reliability) was fundamentally incompatible with cloud-dependent architectures.

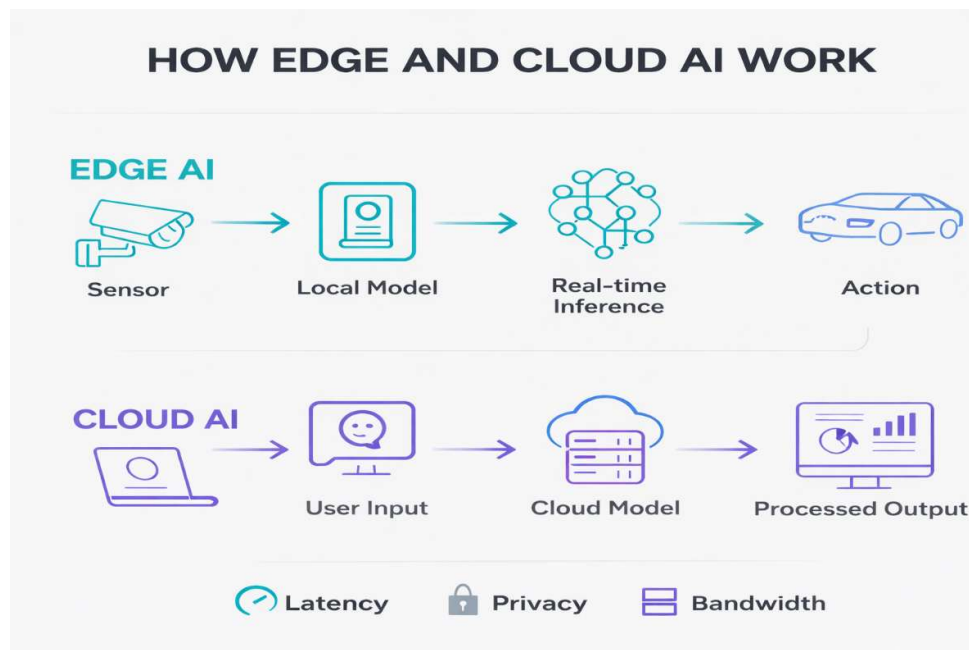


Figure 2.3: Cloud AI vs. Edge AI Pipeline Comparison

The Ai-Thinker VC-02 module, employed as the speech recognition subsystem in the present research, implements Edge AI inference on the Unisound US516P6 processor, a dedicated neural network accelerator designed for embedded speech recognition applications [15]. The module supports configurable command vocabularies of up to 150 entries, operates at 3.3 volts, and communicates recognised command identifiers to host microcontrollers via UART. Of particular significance is the module's support for wake-word-free command recognition, wherein designated high-priority commands trigger immediate system responses without requiring prior wake word activation, a capability exploited in the present design to enable instantaneous emergency alert triggering [15]. Sharif et al. [22] evaluated priority-based task scheduling in edge computing systems for health monitoring, demonstrating that Edge AI architectures could be configured to prioritise emergency-related processing, a paradigm reflected in the present system's firmware architecture.

2.5 GSM-Based Emergency Alert Systems

The Global System for Mobile Communications standard represents the world's most extensively deployed cellular communication infrastructure, providing voice and data services across geographic areas where internet connectivity is unavailable or unreliable [23]. The independence of GSM networks from local power and internet infrastructure renders SMS-based alerting particularly valuable for emergency communication in environments subject to frequent power and network outages.

Ezeofor et al. [23] designed and evaluated a microcontroller-based wireless patient health monitoring system incorporating GSM-based alert transmission, validating the fundamental technical approach of GSM-based medical alerting in resource-constrained environments. The authors noted that SMS delivery offered a significant advantage over internet-based messaging in areas with intermittent connectivity, as SMS messages are queued by the network operator and delivered when network conditions permit. Hameed et al. [24] found that SMS alerts provided a reliable, auditable communication channel for emergency notification that did not require the recipient to be actively monitoring a device at the moment of alert transmission, a significant practical advantage in busy clinical environments.

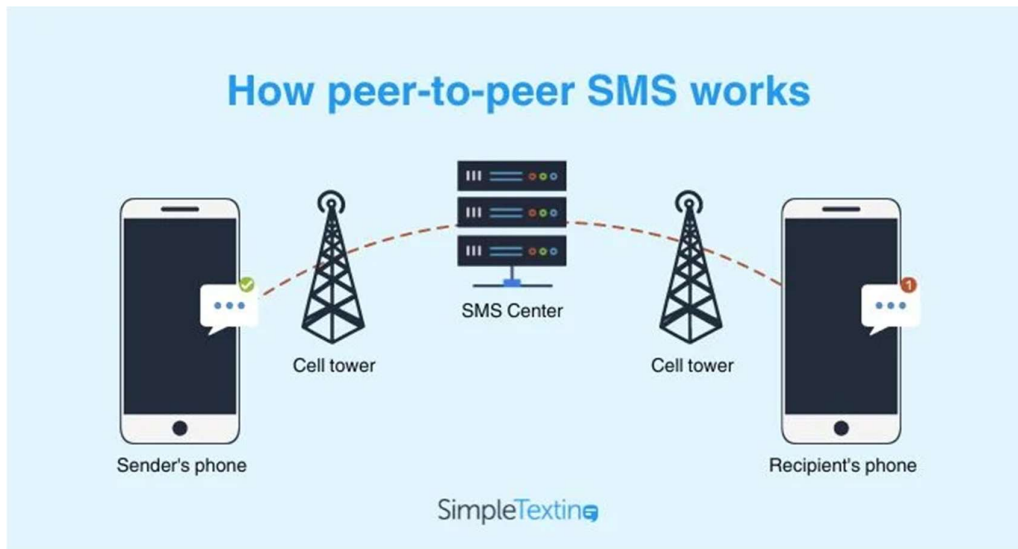


Figure 2.4: How Peer-to-Peer SMS Works

Al-khafajiy et al. [12] developed a comprehensive smart hospital emergency system finding that GSM-based alerting maintained 97.3% delivery reliability during periods when hospital WiFi infrastructure was unavailable, a finding with direct implications for the present research's choice of GSM as the primary emergency communication channel. The SIM800L GSM module provides AT command-based control of SMS transmission over 2G GSM networks [16], with a supply voltage range of 3.4 to 4.4 volts directly compatible with a lithium-ion cell, and peak current consumption of up to 2 amperes during transmission bursts. Shirali-Shahreza [25] demonstrated that SMS provided a resilient alternative communication pathway with lower network resource requirements than voice calls. Jaeyalakshmi et al. [26] reported median SMS delivery latencies of 4.2 seconds under normal network conditions, a benchmark against which the present system's performance is assessed in Chapter Four.

2.6 Nurse Call Systems: Evolution and Limitations

The nurse call system constitutes the foundational patient assistance technology in acute care environments globally [3]. Baballe [27] reviewed wireless nurse call systems, documenting the transition from wired infrastructure to wireless radio-frequency-based systems, while acknowledging that the patient-facing interface (the call button) remained a fundamental accessibility barrier for patients with motor impairments. Aref et al. [28] evaluated wireless nurse call systems across multiple medical institution deployments, finding that wireless systems exhibited vulnerability to radio frequency interference from other hospital electronic equipment and to battery depletion in call button units. The authors recommended the implementation of redundant communication pathways as a critical reliability enhancement.

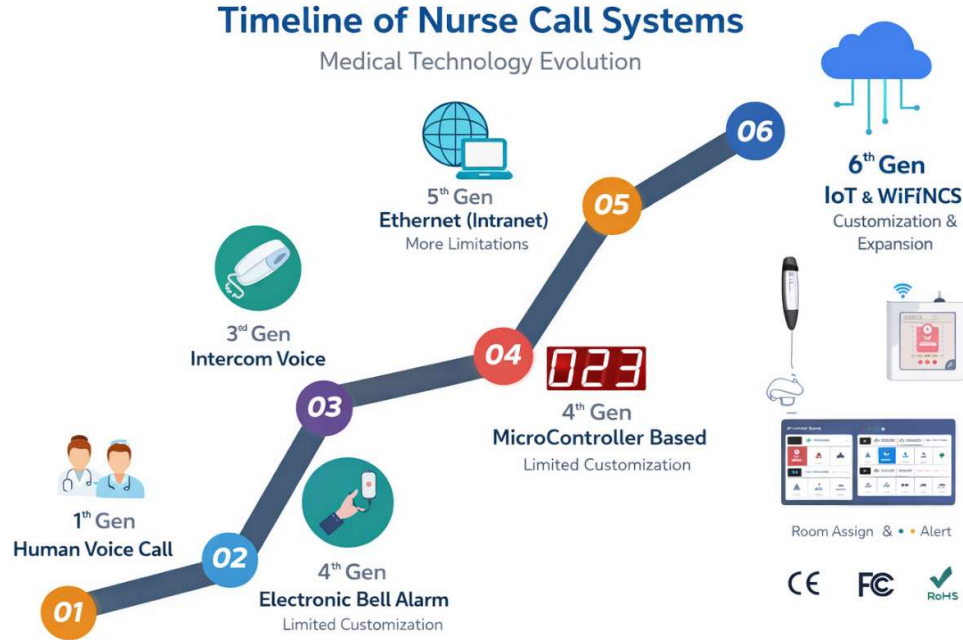


Figure 2.5: Evolution of Nurse Call Systems

Riyad et al. [29] developed an IoT-based nurse calling system utilising local wireless network communication for real-time emergency alerting, demonstrating the feasibility of integrating nurse call functionality with broader hospital IoT infrastructure. However, the system's dependency on local wireless network infrastructure represented a critical single point of failure, as network outages (common in resource-constrained facilities) would render the entire system inoperable. This vulnerability directly motivates the GSM-based alerting architecture of the present research, which operates independently of local network infrastructure.

2.7 Bluetooth Low Energy in Healthcare Applications

Bluetooth Low Energy has emerged as a particularly relevant wireless communication technology for healthcare IoT applications, offering device-to-device communication without internet dependency, energy-efficient operation enabling battery-powered deployment, and a communication range appropriate for intra-room patient assistance applications [30].

Omre and Keeping [30] identified BLE's fundamental suitability for body area network and room-scale healthcare communication, noting that its frequency hopping spread spectrum modulation provided inherent resilience against interference from other wireless devices, a significant advantage in the electromagnetically complex environment of a hospital room. Tosi

et al. [31] conducted a systematic review of BLE performance across forty-three independent evaluation studies, reporting packet reception rates exceeding 95% at distances up to 10 metres under normal indoor conditions, sufficient for reliable communication within the single-room deployment scenario of the present system.

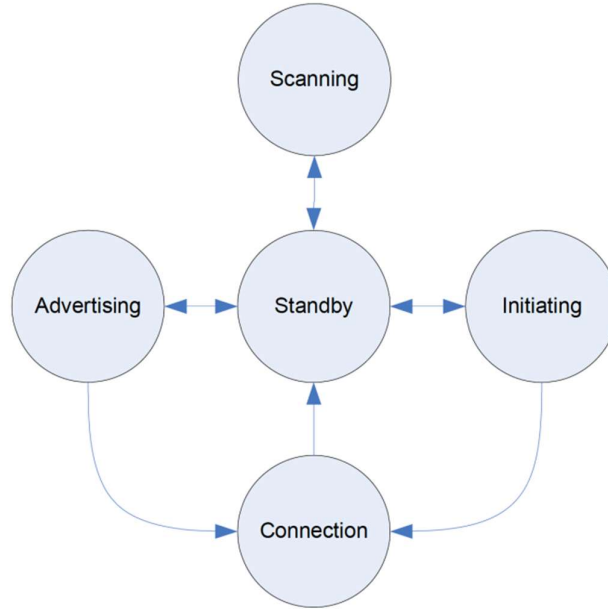


Figure 2.6: BLE Architecture

R. T. D et al. [32] implemented a BLE mesh-based patient monitoring system, demonstrating the scalability of BLE communication architectures for multi-patient hospital deployments. While the present research employs point-to-point rather than mesh BLE communication, the study's validation of BLE reliability in clinical environments informed the BLE implementation approach adopted in the present work. Sherif and Ahmed [33] evaluated a BLE-based real-time location system for medical equipment tracking, providing detailed characterisation of BLE signal propagation through hospital room partitions and in the presence of medical equipment.

2.8 Internet of Things and Home Automation: Foundational Approaches

The relay-based appliance control subsystem of the present research draws on a well-established body of IoT home and building automation literature. Supriya et al. [34] documented relay-switched appliance control as the predominant actuation mechanism for mains-voltage loads in IoT automation systems, identifying the ESP32 microcontroller as an emerging platform of choice due to its integration of WiFi and Bluetooth connectivity, dual-

core processing capability, and extensive peripheral support. Mohamed and Elshenawy [35] demonstrated real-time bidirectional state synchronisation between a central microcontroller and a mobile application interface, directly informing the state synchronisation architecture of the present research.

Koushal et al. [36] demonstrated the ESP32's capability to simultaneously maintain BLE server operation while executing relay switching and serial communication tasks, validating the multi-protocol concurrent operation architecture employed in the present research. Bhaganagare et al. [37] confirmed that ESP32-based voice-controlled automation systems consistently achieved relay switching response times below 500 milliseconds. Maier et al. [38] concluded that the ESP32 demonstrated superior performance-to-cost ratio compared to alternative IoT microcontroller platforms, with particular strengths in concurrent multi-protocol wireless operation. Babiuch and Postulka [39] validated the ESP32 for long-term monitoring applications, and Kalamaras et al. [40] demonstrated its suitability for low-cost IoT implementation in resource-constrained deployment contexts.



Figure 2.7: IoT Home Automation Architecture

2.9 Mobile Applications in Patient Care

Mobile applications have become increasingly integral to healthcare delivery. Flutter, the cross-platform mobile application development framework developed and maintained by Google, has gained significant adoption in healthcare application development due to its single-codebase deployment across Android and iOS platforms, high-performance rendering engine, and extensive third-party library ecosystem [17]. The framework's widget-based architecture facilitates the implementation of large, high-contrast interaction targets appropriate for elderly and mobility-impaired users, a design consideration directly relevant to the patient population targeted by the present research.

Tosi et al. [31] documented that BLE communication between mobile applications and embedded healthcare devices achieved packet reception rates exceeding 95% at distances up to 10 metres under typical indoor conditions, establishing BLE as a reliable channel for mobile application-to-device interaction within the single-room deployment scenario of the present system. A critical software engineering challenge in multi-modal patient assistance systems is the maintenance of consistent device state across concurrent control inputs from multiple modalities. Mohamed and Elshenawy [35] demonstrated that centralised state management on the embedded controller, with propagation to all connected interfaces upon any state change, effectively eliminated state conflicts in multi-modal IoT control systems. This architectural pattern is directly implemented in the present research through a centralised state machine on the ESP32 that notifies the mobile application of every state change regardless of which control modality initiated it.

2.10 Multimodal Interaction Design for Elderly and Mobility-Impaired Users

The multimodal interaction architecture of the present system (integrating voice recognition, mobile application control, and physical switch inputs within a unified state management framework) is grounded in an established body of research demonstrating the superiority of multimodal systems over single-modality alternatives for elderly and mobility-impaired users.

Naumann et al. [13] conducted a foundational investigation of multimodal interaction as a design strategy for inclusive system design, finding that the availability of multiple interaction modalities significantly improved task completion rates and user satisfaction among older users compared to single-modality interfaces. The study identified voice and touch as the most complementary modality pairing for elderly users, noting that each modality compensated for scenarios in which the other was inaccessible, precisely the redundancy rationale underlying

the present system's multimodal architecture. Dias et al. [41] reported significant improvements in successful task completion rates among users with motor impairments when multimodal interaction was available, with the greatest benefits observed among users with the most severe mobility limitations.

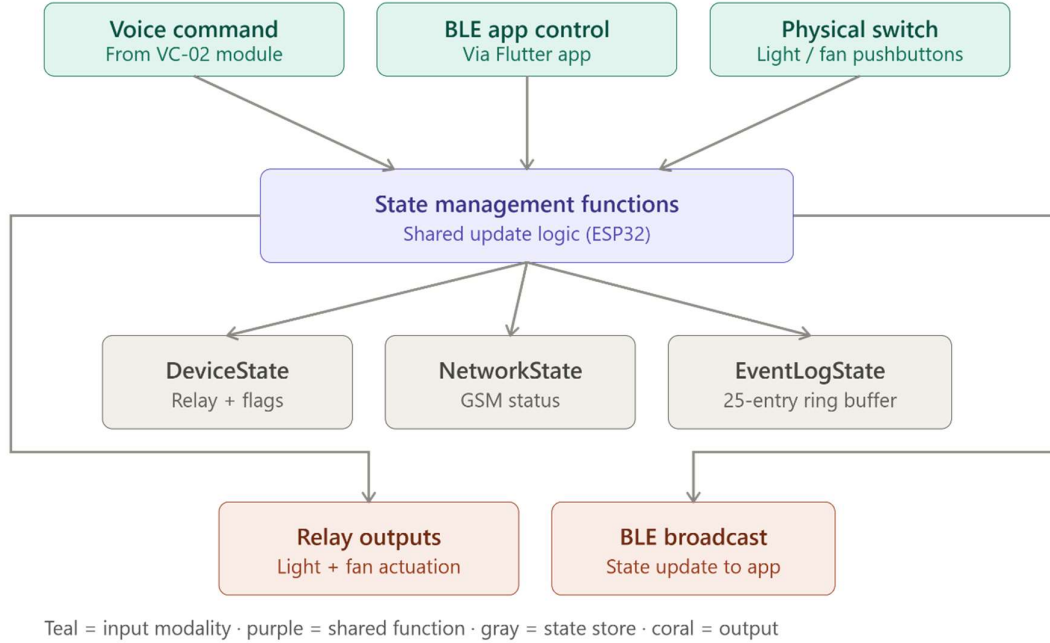


Figure 2.8: Three Control Modalities Converging on ESP32 State Machine

Jian et al. [42] specifically examined the combination of touch and speech interaction modalities for elderly users, finding that the availability of touch-based fallback interaction significantly reduced user anxiety in scenarios where speech recognition failed. This finding directly motivated the inclusion of both mobile application touch control and physical switch interaction as redundant modalities in the present system design. Ismail et al. [43] compared unimodal and multimodal interaction approaches for elderly users, finding that multimodal interfaces achieved significantly higher first-attempt success rates and lower error rates across all age groups.

2.11 Healthcare Technology Adoption in Nigeria

The specific deployment context of the present research (Nigerian healthcare facilities) introduces considerations that distinguish it from the predominantly Western and Asian contexts in which most related works have been conducted. Oluwale et al. [5] documented a strong positive correlation between infrastructure reliability and measurable healthcare quality

indicators in Nigerian facilities, identifying unreliable power supply as the most pervasive infrastructure constraint, affecting 78% of surveyed facilities, followed by inadequate internet connectivity affecting 64%, findings that directly validate the infrastructure-independent design philosophy of the present system.

Umar et al. [6] identified mobile technology and embedded IoT systems as the most promising near-term technological interventions for Nigerian healthcare, specifically noting that solutions requiring reliable internet connectivity would necessarily exclude the majority of Nigerian primary and secondary healthcare facilities. Kanoute [44] documented specific challenges associated with health technology adoption in Northern Nigeria (the geographic region in which the present research is conducted), identifying electricity supply instability, limited technical maintenance capacity, and varying levels of digital literacy among clinical staff as key determinants of technology adoption success. These contextual factors informed several design decisions in the present research, including the prioritisation of simple, robust hardware components and the implementation of a physical switch fallback modality that remains operational without any electronic intelligence.

2.12 Research Gap

The foregoing review of literature across ten thematic domains reveals a consistent and significant gap at the intersection of offline Edge AI speech recognition, GSM-based emergency alerting, multimodal patient assistance, mobile application integration, and resource-constrained healthcare deployment.

Existing voice-controlled patient assistance systems are uniformly dependent on cloud-based speech processing infrastructure, rendering them inoperable in the absence of internet connectivity [11][10][19]. GSM-based nurse alerting systems, while demonstrating appropriate infrastructure independence, require physical button activation inaccessible to mobility-impaired patients [23][24]. Multimodal assistive systems, where documented, have been designed exclusively for well-resourced environments with reliable connectivity [13][41][42]. IoT-based home automation systems demonstrate the technical viability of relay-based appliance control through voice and mobile application interfaces but are not designed for clinical deployment and lack emergency alerting functionality [34][35][37].

Critically, no prior work identified in this review has demonstrated the integration of fully offline Edge AI speech recognition with no cloud dependency, multi-phrase culturally adaptive

command vocabularies, wake-word-free emergency command detection, GSM-based SMS emergency alerting independent of hospital network infrastructure, a dedicated companion mobile application with BLE redundant control and real-time bidirectional state synchronisation, physical switch redundancy, unified centralised state management, and relay-based mains-voltage appliance control, all within a design specifically optimised for resource-constrained healthcare environments characterised by power instability and internet unavailability. The present research directly addresses this gap.

2.13 Conclusion

This chapter has reviewed the theoretical and empirical foundations underpinning the present research across ten thematic domains. The review has established the clinical significance of accessible nurse call and patient assistance systems for mobility-impaired hospital patients, the technical maturity of Edge AI for offline speech recognition in embedded systems, the reliability of GSM-based SMS communication for emergency alerting in resource-constrained environments, the performance characteristics of Bluetooth Low Energy for intra-room healthcare communication, the established design patterns of IoT relay-based appliance control, the value of mobile applications as a redundant control modality, the demonstrated superiority of multimodal interaction architectures for elderly and mobility-impaired users, and the specific infrastructure constraints characterising Nigerian healthcare facilities. A comprehensive research gap has been identified across the convergence of these domains. The following chapter presents the complete system design and methodology, detailing the hardware component selection and justification, circuit design, firmware architecture, and design calculations that form the technical foundation of the proposed system.

CHAPTER THREE

SYSTEM DESIGN AND METHODOLOGY

3.1 Introduction

This chapter presents the complete design and methodology of the Multimodal Offline Voice-Controlled Patient Assistance System with GSM Emergency Alerting. The chapter covers the overall system architecture, hardware component selection with justification over alternatives, complete circuit design, firmware and mobile application architecture, and eleven design calculations that validate every major component selection and circuit parameter. All design decisions are grounded in the operational requirements identified in Chapter Two: complete offline operation, GSM-independent emergency alerting, multi-modal redundant control, relay-based mains-voltage appliance switching, and suitability for deployment in resource-constrained healthcare environments.

3.2 System Architecture and Block Diagram

The system architecture follows a centralised hub-and-spoke model in which the ESP32 microcontroller serves as the sole processing and state management node. All inputs from the three control modalities are routed to the ESP32, which processes commands, maintains the authoritative system state, drives all outputs, and manages communication with both the GSM module and the mobile application. This centralised architecture eliminates the possibility of state conflicts between modalities and simplifies fault diagnosis [35]. The system is organised into five functional units (the Power Unit, the Processing Unit, the Input Unit, the Communication Unit, and the Output Unit), corresponding directly to the sections of the schematic. Figure 3.1 shows the block diagram illustrating the relationship between these units.

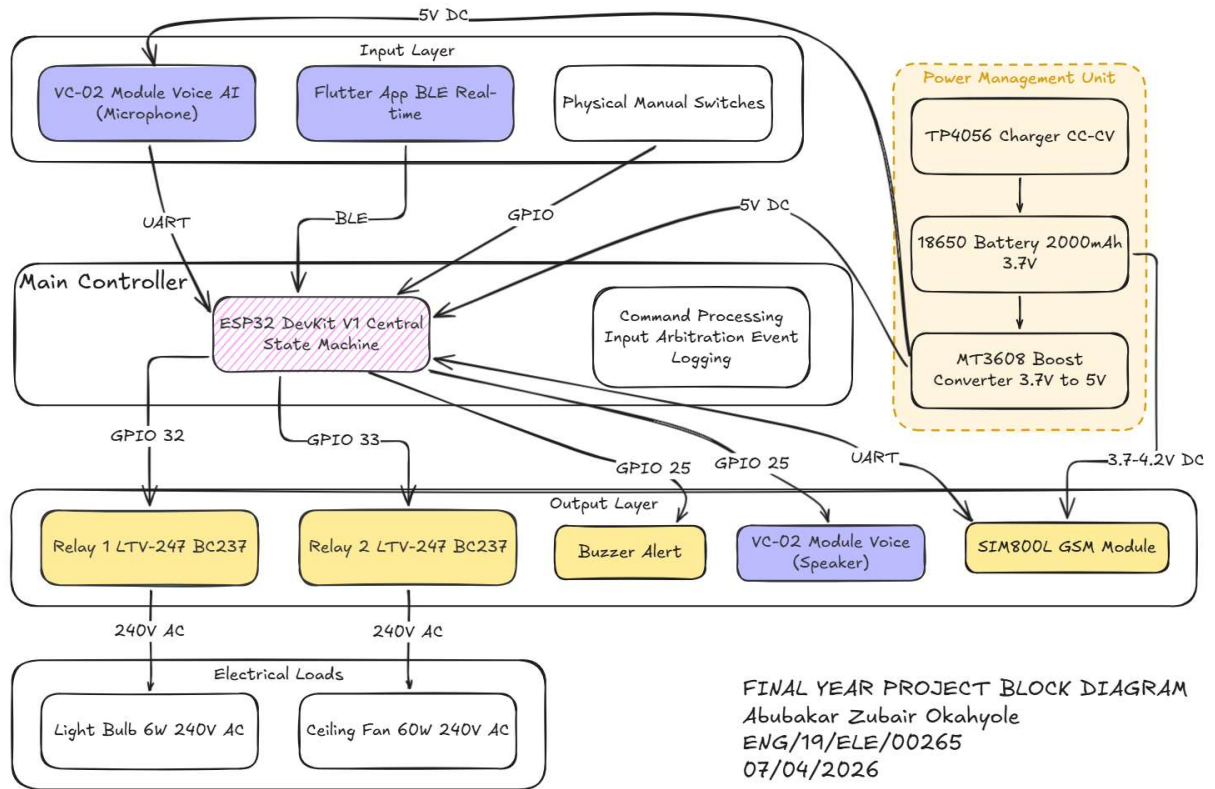


Figure 3.1: System Architecture Block Diagram

The Power Unit supplies regulated voltages to all other units. The 18650 lithium-ion battery is charged by the TP4056 module; the battery output after the main power switch feeds two parallel branches: the SIM800L GSM module receives the battery voltage directly, while the MT3608 boost converter steps the battery voltage up to a stable 5V rail that powers all remaining components. The Processing Unit contains the ESP32 DevKit V1 microcontroller and the VC-02 Edge AI voice module, coordinating all subsystems via UART and GPIO. The Input Unit provides two physical pushbutton switches and the condenser microphone connected to the VC-02. The Communication Unit contains the SIM800L GSM module and antenna, transmitting SMS emergency alerts over the GSM network. The Output Unit contains two identical relay driver sub-circuits (each comprising an LTV-247 optocoupler and BC237 NPN transistor driving an SRD-05VDC-SL-C relay), switching the 240V AC mains supply to the room light and ventilation fan, together with the active buzzer and speaker.

3.3 Hardware Component Selection and Justification

A key engineering requirement of this project is that every component selection must be justified against alternatives. This section presents the rationale for each major component chosen for the system.

3.3.1 Microcontroller: ESP32 DevKit V1

The ESP32 DevKit V1, based on the ESP32-WROOM-32 module, was selected as the central processing unit. The module integrates a dual-core Xtensa LX6 processor at up to 240MHz, 520kB of SRAM, 4MB of flash, and hardware support for Bluetooth 4.2 including BLE [45]. The selection of the ESP32 over alternatives including the Arduino Mega 2560 was justified on four technical grounds. First, the system requires two simultaneous hardware UART channels (one for the VC-02 and one for the SIM800L); the Arduino Uno provides only one hardware UART and would require unreliable software serial for the second channel. Second, the ESP32 integrates a BLE radio on-chip, eliminating the need for an external HC-05 or HC-06 module that would consume an additional UART port. Third, the ESP32's 240MHz dual-core processor provides sufficient computational resources to manage concurrent BLE communication, voice command polling, GSM management, and GPIO monitoring simultaneously without timing conflicts [38]. Fourth, the ESP32 provides Non-Volatile Storage (NVS) for persistent configuration, allowing nurse contact numbers and patient bed identifiers to survive power cycles without external EEPROM hardware.

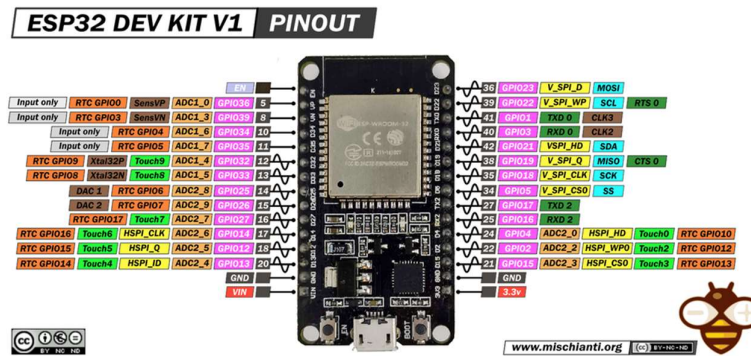


Figure 3.2: ESP32 DevKit V1 Annotated Pinout Diagram

3.3.2 Voice Recognition Module: VC-02

The Ai-Thinker VC-02 module implements offline Edge AI speech recognition on the Unisound US516P6 neural network processor [15]. The VC-02 was selected over cloud-based voice recognition (Amazon Alexa, Google Speech API) because cloud platforms require stable

internet connectivity, which was unavailable in the target deployment environment. It was also selected over CMU Sphinx running on a Raspberry Pi: Sphinx on a Raspberry Pi requires 5V at 700mA, a Linux operating system with a 30-second boot time, and significantly higher cost, while the VC-02 operates at 3.3V consuming approximately 50mA, boots instantaneously as an embedded peripheral, and supports a configurable vocabulary of up to 150 command phrases. Critically, the VC-02 supports wake-word-free command designation, meaning designated emergency commands are recognised and acted upon immediately without requiring a preceding wake word. This property is clinically essential: a patient in acute distress cannot be expected to first say a wake word before stating their emergency [15].

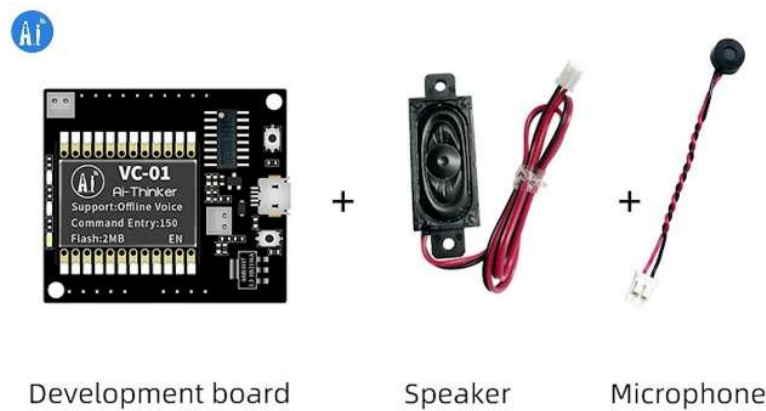


Figure 3.3: VC-02 Development Board, Speaker and Microphone Pinout Diagram

3.3.3 GSM Communication Module: SIM800L

The SIM800L was selected over WiFi-based alert transmission on the basis of infrastructure independence. WiFi alerting requires reliable hospital network infrastructure; when the hospital network fails during a power outage (a common occurrence in Nigerian facilities), all WiFi-dependent alerts fail simultaneously. The SIM800L operates over the GSM cellular network, which maintains independent infrastructure, battery-backed base stations, and near-complete coverage across Nigeria on the 900MHz and 1800MHz bands used by Airtel Nigeria [16]. SMS messages are network-queued and delivered when conditions permit, providing a fundamentally more reliable emergency communication channel than WiFi in the target environment. The SIM800L's supply voltage range of 3.4V to 4.4V is directly compatible with the 18650 battery's discharge curve, permitting direct connection without a voltage regulator on the SIM800L power path.

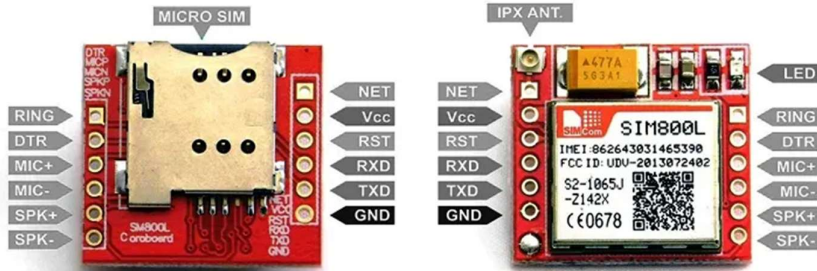


Figure 3.4: SIM800L Annotated Pinout Diagram

3.3.4 Relay Driver Circuit: LTV-247 Optocoupler and BC237 Transistor

The relay driver circuit uses an LTV-247 optocoupler in series with a BC237 NPN transistor. This two-stage design was chosen over a single-transistor relay driver for two reasons. First, the optocoupler provides complete galvanic isolation between the ESP32 GPIO (3.3V logic, sensitive CMOS) and the relay coil circuit (5V, inductive load). When a relay coil is de-energised, it generates a back-EMF voltage spike that can damage unprotected CMOS inputs; the optocoupler prevents this spike from reaching the ESP32 regardless of the flyback diode performance. Second, the relay coil operates at 5V from the VCC rail while the ESP32 GPIO outputs 3.3V; the optocoupler accepts the 3.3V input on its LED side while its phototransistor output drives the BC237 from the 5V rail, eliminating any level-shifting requirement. The BC237 was selected for its low saturation voltage ($V_{CE(sat)} = 0.2V$), adequate current gain (h_{FE} minimum 100), and availability. Flyback diodes D4 and D5 are placed anti-parallel across each relay coil to suppress back-EMF spikes during relay de-energisation [46].

3.3.5 Relay: SRD-05VDC-SL-C

The SRD-05VDC-SL-C relay was selected for its 5V coil rating matching the VCC rail, 70 Ohm coil resistance, and 10A/250VAC contact rating. The 10A contact rating provides a safety margin of 400x over the 25mA light load and 40x over the 250mA fan load, ensuring the contacts are not stressed by the connected AC loads and providing substantial headroom for motor and lamp inrush currents at start-up. This selection is validated quantitatively in Section 3.8.

3.3.6 Boost Converter: MT3608

The MT3608 boost converter was selected to step the battery voltage (3.7V nominal, 3.4V minimum) up to the stable 5V required by the ESP32 VIN pin, relay module, and buzzer. Powering the ESP32 from USB permanently was rejected because it creates infrastructure

dependency, requiring a USB power source in the hospital room at all times. The MT3608 was chosen over the XL6009 boost converter for its smaller package, lower minimum input voltage (2V vs 3V), and higher switching frequency (1.2MHz) allowing smaller external components. Its 4A internal switch current limit provides more than 10x headroom over the maximum load current of 383mA on the 5V rail [47].

3.3.7 Battery Charger: TP4056

The TP4056 implements the industry-standard constant-current constant-voltage (CC-CV) charging profile, automatic charge termination at 4.2V, and overcurrent protection [47]. Its maximum charge current is configurable via a single programming resistor. The TP4056 provides charge status indication via the CHRG and STDBY pins connected to indicator LEDs in the Power Unit.

3.3.8 Power Switch

A momentary pushbutton POWER_SWITCH is placed between the TP4056 output terminals and both the MT3608 input and the SIM800L VCC supply. This allows complete system power-down while the battery remains connected to the TP4056 for charging. Both the SIM800L and MT3608 are on the switched side of this switch, ensuring complete system shutdown when the switch is open.

3.4 Complete Circuit Design

The complete circuit is implemented across the five unit sections of the KiCad schematic shown in Figure 3.5. The circuit design is described below unit by unit following the schematic organisation.

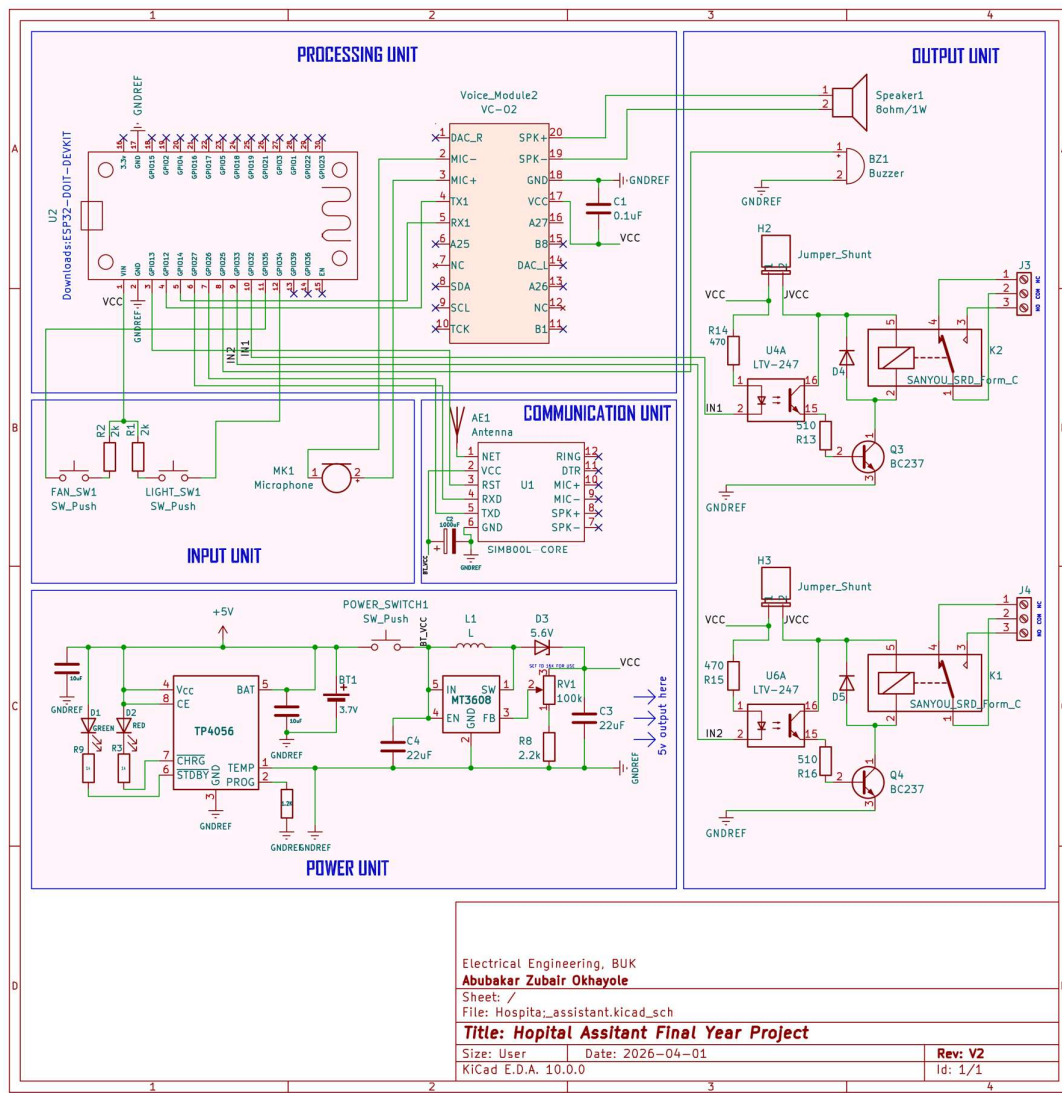


Figure 3.5: Complete System Circuit Diagram (KiCad)

3.4.1 Power Unit

The 18650 battery connects to the BAT+ and BAT- terminals of the TP4056. The module's output terminals (OUT+ and OUT-) supply the power rail through the POWER_SWITCH. Decoupling capacitors of 10 μ F are placed on the TP4056 VCC and BAT lines. The MT3608 boost converter receives the switched battery voltage at its IN pin. The output voltage is set by the feedback resistor network, comprising potentiometer RV1 (100k Ω) and resistor R8 (2.2k Ω), connected between the output and the FB pin. Inductor L1 and Schottky diode D3 (5.6V rated) form the boost converter's energy storage and rectification elements. Output capacitors C3 and C4 (22 μ F each) filter the switching ripple. The regulated 5V output is

labelled VCC throughout the schematic. The SIM800L VCC pin connects directly to the switched battery positive rail, bypassing the MT3608. A 1000 μ F electrolytic capacitor is placed directly across the SIM800L VCC and GND pins to supply the module's 2A peak transient demand during GSM transmission bursts.

3.4.2 Processing Unit

The ESP32 DevKit V1 receives its supply at the VIN pin from the 5V VCC rail, providing the 3.3V output via its onboard AMS1117 regulator for the ESP32 core and the VC-02 module. The VC-02 VCC pin (pin 17) connects to the ESP32 3.3V output. Communication between the two is via UART: the VC-02 TX1 pin (pin 4) connects to ESP32 GPIO12 (UART1 RX) and the VC-02 RX1 pin (pin 5) connects to ESP32 GPIO14 (UART1 TX). A 0.1 μ F decoupling capacitor C1 is placed on the VC-02 VCC pin. The VC-02 SPK+ and SPK- pins connect to an 8Ohm/1W speaker for voice response playback; the MIC+ and MIC- pins connect to microphone MK1 in the Input Unit.

3.4.3 Input Unit

Two pushbutton switches, FAN_SW1 and LIGHT_SW1, are connected between GPIO35 and GPIO34 respectively and ground. Pull-up resistors R1 and R2 (2k Ω each) connect these GPIO pins to the 3.3V rail. GPIO 34 and 35 are input-only pins on the ESP32 with no internal pull-up capability [45], making external pull-up resistors mandatory. The switch inputs and pull-up resistor values are validated in Calculation 3.8.9.

3.4.4 Communication Unit

The SIM800L RXD pin (pin 4) connects to ESP32 GPIO27 (UART2 TX) and the TXD pin (pin 5) connects to ESP32 GPIO26 (UART2 RX), forming a dedicated UART channel at 9600 baud for AT command communication. The antenna connector AE1 connects to the GSM antenna AE1. The RST pin connects to ESP32 GPIO13, enabling firmware-controlled hardware reset of the SIM800L via the RESET_GSM BLE command. Table 3.1 presents the complete GPIO pin assignment.

Table 3.1: ESP32 GPIO Pin Assignment

GPIO Pin	Function	Direction	Connected Component
GPIO 12	VC-02 UART RX	Input	VC-02 TX1 pin

GPIO 13	SIM800L RST	Output	SIM800L RST pin
GPIO 14	VC-02 UART TX	Output	VC-02 RX1 pin
GPIO 25	Buzzer	Output	Active Buzzer (+)
GPIO 26	SIM800L UART RX	Input	SIM800L TXD pin
GPIO 27	SIM800L UART TX	Output	SIM800L RXD pin
GPIO 32	Relay 1: Light	Output	LTV-247 U4A pin 1 (anode via R14)
GPIO 33	Relay 2: Fan	Output	LTV-247 U6A pin 1 (anode via R15)
GPIO 34	Light Switch Input	Input	SW LIGHT_SW1 + 2k Ω pull-up to 3.3V
GPIO 35	Fan Switch Input	Input	SW FAN_SW1 + 2k Ω pull-up to 3.3V

3.4.5 Output Unit: Relay Driver Circuit

Each relay driver sub-circuit operates identically for the light (RELAY1) and fan (RELAY2) channels. The circuit is shown in Figure 3.6. The ESP32 GPIO output (GPIO32 for RELAY1, GPIO33 for RELAY2) drives the anode of the LTV-247 optocoupler's internal LED through a 470 Ω current-limiting resistor (R14 for RELAY1, R15 for RELAY2). The optocoupler's phototransistor output drives the base of transistor Q3 or Q4 (BC237) through resistor R13 or R16 (510 Ω each). The BC237 collector connects to one end of the relay coil; the other end connects to VCC via the Jumper_Shunt (H2/H3) connecting JVCC to VCC. A flyback diode (D4/D5) is placed anti-parallel across each relay coil to suppress the back-EMF spike on de-energisation. The relay's Normally Open (NO) and Common (COM) contacts are brought out to connectors J3 and J4; when a relay is energised, the NO contact closes, completing the 240V AC circuit to the connected appliance. The active buzzer BZ1 connects between GPIO25 and ground, driven directly from the ESP32 GPIO.

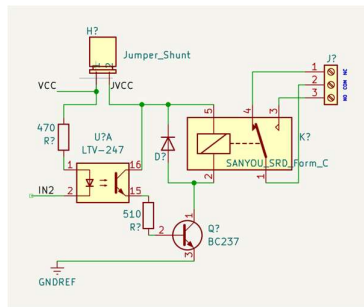


Figure 3.6: Relay Driver Sub-Circuit (Single Channel)

3.5 Firmware Architecture

3.5.1 Overview

The ESP32 firmware is implemented within the Arduino framework using a modular header-file architecture comprising eight files. The runtime model is a cooperative polling scheduler; no RTOS is used. Each module's loop function executes and returns promptly, ensuring no single subsystem monopolises the processor. Table 3.2 summarises the module responsibilities.

Table 3.2: Firmware Module Architecture (v4.0)

Module File	Primary Responsibility
config.h	Pin definitions, BLE UUIDs, command codes, timing constants, default contact configuration
state.h	DeviceState, NetworkState, and EventLogState data structures and shared service prototypes
relay_module.h	Relay and buzzer control; non-blocking buzzer state machine (2s beeps, 3 repetitions)
sim800_module.h	GSM initialisation, periodic network polling, AT command transmission, SMS with up to 3 retries
vc02_module.h	Two-byte UART command reception from VC-02, semantic decoding, emergency and nurse-call workflows
switch_module.h	Pushbutton debounce with 50ms stable-state logic and auto-detected press polarity per switch
ble_module.h	BLE GATT server, ACK/NACK protocol, state broadcast, GET_LOG and RESET_GSM command handling
Hospital_Assistant_V4.ino	NVS persistent config load/save, 25-entry ring-buffer event log, main cooperative loop

3.5.2 State Machine Design

The centralised state machine maintains three state structures: DeviceState (relay states, emergency flag, nurse call flag, last command source), NetworkState (GSM registration status, timestamp of last network check), and EventLogState (25-entry ring buffer of all system events). All three control modalities (voice, BLE application, and physical switch) update state exclusively through defined state management functions, which then drive the relay hardware outputs and broadcast the updated state to the mobile application via BLE notification. This architecture guarantees that the device state is always consistent across all connected interfaces regardless of which modality initiated the most recent change.

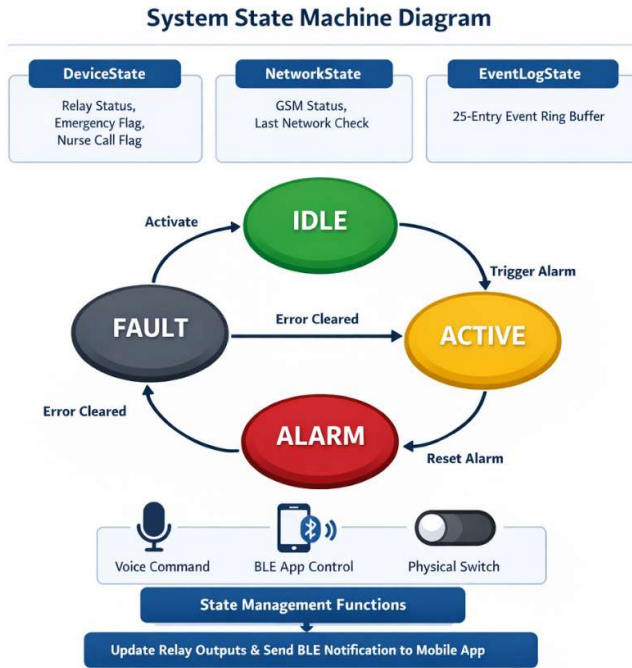


Figure 3.7: System State Machine Diagram

3.5.3 Voice Command Processing

Voice commands are received from the VC-02 as two-byte sequences over the UART1 hardware serial interface. The firmware reads the first byte and immediately awaits the second byte with a 50ms timeout, forming a 16-bit command identifier by concatenating the two bytes. The assembled identifier is matched against the defined command table and the corresponding state management function invoked. Emergency and nurse-call commands are designated wake-word-free, enabling immediate recognition without prior wake word activation [15]. Table 3.3 presents the complete voice command mapping.

Table 3.3: Voice Command Mapping

Command Category	Hex Identifier	Sample Recognised Phrases	Wake-Word Free
Wake Word	0xAA00	"Hello smart nurse", "Hi pudding", "Ni hao"	No
Emergency Alert	0xAA01	"I need help", "This is an emergency"	Yes
Call Nurse	0xAA02	"Call the nurse", "Help me call nurse"	No
Light ON	0xAB11	"Turn on the light", "Light on", "I need light"	No
Light OFF	0xAB12	"Turn off the light", "Light off", "I no need light"	No
Fan ON	0xFF11	"Turn on the fan", "Fan on", "I need fan"	No
Fan OFF	0xFF12	"Turn off the fan", "Fan off", "Off the fan"	No

3.5.4 BLE Communication Protocol

BLE communication is implemented using the ESP32's integrated BLE hardware, configured as a GATT server. Every command received from the mobile application returns an ACK or NACK response, eliminating silent command failures. The RESET_GSM command pulls the SIM800L RST pin (GPIO13) LOW for 100ms then HIGH, clearing the network ready flag and re-initiating GSM registration. The GET_LOG command returns the 25 most recent event log entries prefixed with "LOG:" for detection by the mobile application. State updates are broadcast to the connected application in the structured format: LIGHT:[0/1], FAN:[0/1], EMERGENCY:[0/1], NETWORK:[0/1], SOURCE:[voice/app/switch].

3.5.5 Persistent Configuration and Event Logging

The v4.0 firmware implements NVS-based persistent configuration storage enabling the patient bed identifier, nurse phone number, and doctor phone number to survive power cycles without re-entry. Configuration values are written to NVS upon receipt of SET_BED:, SET_NURSE:, and SET_DOCTOR: BLE commands and loaded from NVS at system startup before BLE initialisation. The 25-entry ring-buffer event log records all system events (voice commands, BLE commands, switch activations, and emergency triggers) with correct overflow behaviour ensuring the 25 most recent events are always available for retrieval.

3.6 Voice Command Configuration

The VC-02 module's command vocabulary was configured through the Ai-Thinker cloud configuration platform, which compiled the specified vocabulary into a binary firmware image for flashing to the module [15]. The configuration designated emergency and nurse-call commands as wake-word-free, specified multiple English phrase variants per command category to accommodate the natural speech variation of Nigerian patients, and assigned the two-byte hexadecimal command identifiers documented in Table 3.3. The complete vocabulary including all phrase variants is documented in Appendix C. The compiled binary firmware was flashed to the VC-02 module via a USB-to-TTL serial adapter.

3.7 Mobile Application Design

3.7.1 Application Architecture

The companion mobile application was developed using the Flutter framework version 3.x [17], targeting the Android platform. The application architecture follows a layered structure: screens, services, models, widgets, and utilities. BLE communication is implemented using the flutter_blue_plus library [48]. Table 3.4 summarises the four application screens.

Table 3.4: Mobile Application Screen Structure

Screen	Function
Splash Screen	Application startup with animated branding display and custom microphone-heartbeat icon
Home Screen	Primary control interface with BLE connection management, relay toggles, real-time network and device status, and 3-dot menu (Event Log, Reset GSM, Settings, About, Disconnect)
Settings Screen	Configuration of patient bed number, nurse and doctor emergency contact numbers, and ESP32 device name
About Screen	Developer information, university affiliation, project details, technology stack, and contact links

3.7.2 State Synchronisation and BLE Protocol

Upon establishing a BLE connection, the application immediately transmits GET_STATE to the ESP32, which responds with the full current device state. Thereafter, the ESP32 sends BLE notifications on every state change regardless of source modality, keeping the application display synchronised at all times. Settings values are stored locally using shared_preferences and transmitted to the ESP32 via BLE SET_BED:, SET_NURSE:, and SET_DOCTOR: commands, where they are written to NVS for persistence across power cycles.

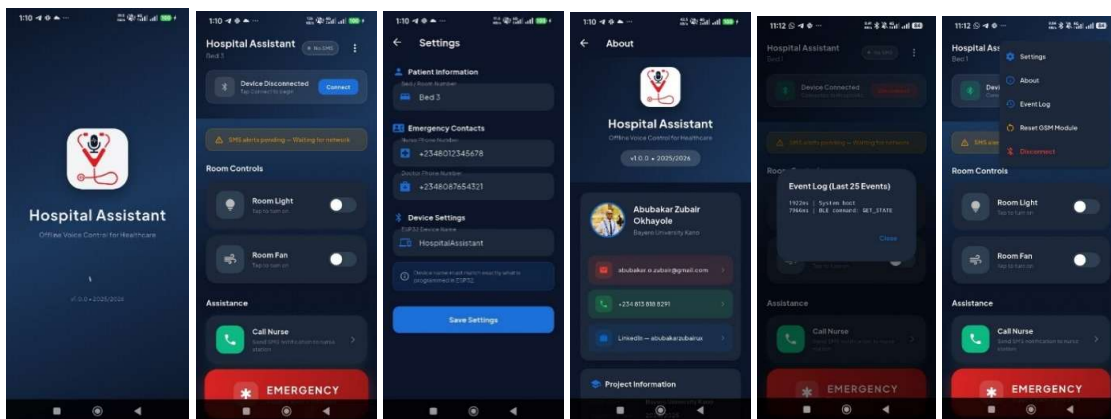


Figure 3.8: Mobile Application Screen Layouts

3.8 Design Calculations

The following eleven calculations validate the key design parameters of the circuit implemented in the KiCad schematic. All component values used are as marked in the schematic and as specified in the relevant datasheets [45][46][47].

3.8.1 MT3608 Output Voltage Setting

The MT3608 regulates its output using an internal reference voltage of $V_{ref} = 0.6V$. The output voltage is set by the feedback resistor divider connected to the FB pin, given by Equation (3.1):

$$V_{out} = V_{ref} \times \left(1 + \frac{R_1}{R_2}\right) \quad (3.1)$$

To set $V_{out} = 5V$, the required RV1 setting is:

$$\begin{aligned} RV_1 &= \left(\frac{V_{out}}{V_{ref}} - 1\right) \times R_8 \\ &= \left(\frac{5}{0.6} - 1\right) \times 2200 \\ &= 7.333 \times 2200 \\ &= 16,133 \Omega \approx 16.1 k\Omega \end{aligned} \quad (3.2)$$

The 100k Ω potentiometer RV1 is adjusted to 16.1% of its total travel to achieve 5V output.

Verification: $V_{out} = 0.6 \times (1 + 16133/2200) = 0.6 \times 8.333 = 5.0V$

3.8.2 MT3608 Boost Converter Duty Cycle

The theoretical duty cycle in continuous conduction mode is given by Equation (3.3):

$$D = 1 - \left(\frac{V_{in}}{V_{out}}\right) \quad (3.3)$$

At nominal battery voltage ($V_{in} = 3.7V$): $D = 1 - (3.7/5.0) = 0.26$ (26%). At minimum battery voltage ($V_{in} = 3.4V$): $D = 1 - (3.4/5.0) = 0.32$ (32%). Both values are well within the MT3608's operational range, confirming stable boost conversion across the full battery discharge curve.

3.8.3 MT3608 Maximum Output Current and Load Verification

The maximum continuous load on the 5V VCC rail, with both relays energised and the buzzer active, is calculated as the sum of all connected loads:

Table 3.5: System Power Budget (5V VCC Rail)

Component	Current Estimate	Source
ESP32 DevKit V1 (active, BLE enabled)	160 mA	[45]
VC-02 Edge AI Module (active listening)	50 mA	[15]
RELAY1 coil (when energised)	71.4 mA	Calculated (see Section 3.8.5)
RELAY2 coil (when energised)	71.4 mA	Calculated (see Section 3.8.5)
Active Buzzer BZ1	30 mA	Typical
Total maximum (both relays + buzzer active)	382.8 mA	Calculated

MT3608 rated continuous output current: 2000 mA. Current headroom factor = $2000 / 382.8 = 5.2\times$. The MT3608 is confirmed to be not stressed under any system operating condition.

3.8.4 MT3608 Input Current from Battery

Using power conservation with MT3608 efficiency $\eta = 85\%$, the battery input current is given by Equation (3.4):

$$I_{in} = \frac{V_{out} \times I_{out}}{V_{in} \times \eta} \quad (3.4)$$

At maximum load ($I_{out} = 382.8 \text{ mA}$, $V_{in} = 3.7\text{V}$): $I_{in} = (5 \times 0.3828) / (3.7 \times 0.85) = 1.914 / 3.145 = 0.609 \text{ A}$.

At idle (both relays off, no buzzer, $I_{out} = 210 \text{ mA}$): $I_{in} = (5 \times 0.210) / (3.7 \times 0.85) = 1.05 / 3.145 = 0.334 \text{ A}$.

3.8.5 Relay Coil Current

The SRD-05VDC-SL-C relay has a rated coil voltage of 5V and a coil resistance of 70 Ohm. The energisation current is:

$$I_{coil} = \frac{V_{coil}}{R_{coil}} = \frac{5}{70} = 71.4 \text{ mA} \quad (3.5)$$

The relay's minimum pull-in voltage is 3.75V (75% of rated). Since the MT3608 maintains a regulated 5V output continuously, the relay receives its full rated voltage at all times regardless of battery state of charge, ensuring reliable pull-in.

3.8.6 BC237 Transistor Saturation Analysis

Parameters from datasheet: $V_{BE} = 0.7V$, $V_{CE(sat)} = 0.2V$, $h_{FE(min)} = 100$. Collector current: $I_C = I_{coil} = 71.4 \text{ mA}$.

Minimum base current required for saturation:

$$I_{B(min)} = \frac{I_C}{h_{FE}} = \frac{71.4}{100} = 0.714 \text{ mA} \quad (3.6)$$

Actual base current supplied through $R_{13} = R_{16} = 510 \text{ Ohm}$ from the 5V rail:

$$\begin{aligned} I_{B(actual)} &= \frac{(V_{CC} - V_{BE})}{R_B} = \frac{(5 - 0.7)}{510} \\ &= \frac{4.3}{510} = 8.43 \text{ mA} \end{aligned} \quad (3.7)$$

Saturation factor:

$$SF = \frac{I_{B(actual)}}{I_{B(min)}} = \frac{8.43}{0.714} = 11.8 \quad (3.8)$$

A saturation factor exceeding 5 guarantees deep saturation of the BC237, ensuring the relay closes reliably with minimal $V_{CE(sat)}$ power dissipation in the transistor.

3.8.7 LTV-247 Optocoupler LED Drive Current

The LTV-247 forward voltage is $V_F = 1.2V$. With $R_{14} = R_{15} = 470 \text{ Ohm}$ and ESP32 GPIO output of 3.3V:

$$\begin{aligned} I_F &= \frac{(V_{GPIO} - V_F)}{R} = \frac{(3.3 - 1.2)}{470} \\ &= \frac{2.1}{470} = 4.47 \text{ mA} \end{aligned} \quad (3.9)$$

This is within the LTV-247 specified I_F range of 1mA to 60mA. At $I_F = 4.47\text{mA}$ the Current Transfer Ratio (CTR) is approximately 80%, yielding sufficient phototransistor output to drive the BC237 base through R_{13}/R_{16} . The GPIO source current of 4.47mA is well within the ESP32's 40mA maximum per GPIO pin, confirming safe operation.

3.8.8 AC Load Current and Relay Contact Safety Margin

Nigerian mains supply: 240V AC, 50Hz. Load currents:

$$I_{bulb} = \frac{P}{V} = \frac{6\text{ W}}{240\text{ V}} = 25\text{ mA} \quad (3.10)$$

$$I_{fan} = \frac{P}{V} = \frac{60\text{ W}}{240\text{ V}} = 250\text{ mA} \quad (3.11)$$

SRD-05VDC-SL-C contact rating: 10A / 250V AC.

Safety margins: Light circuit = $10 / 0.025 = 400\times$. Fan circuit = $10 / 0.250 = 40\times$. Fan motor inrush current (worst case $10\times$ steady state): $2.5\text{A} \ll 10\text{A}$ rated . These margins confirm the relay contacts will not be stressed by the connected loads under any operating condition including start-up inrush.

3.8.9 External Pull-up Resistor for GPIO 34 and GPIO 35

GPIO 34 and 35 are input-only pins with no internal pull-up resistors [45]. External resistors $R1 = R2 = 2\text{k}\Omega$ connect each pin to the 3.3V rail. Pull-up current:

$$I_{pullup} = \frac{V}{R} = \frac{3.3}{2000} = 1.65\text{ mA} \quad (3.12)$$

This is below the ESP32's 40mA GPIO limit and provides sufficient current to maintain a defined logic HIGH at the input pin when the switch is open, preventing floating input conditions that would cause false triggering.

3.8.10 SIM800L Decoupling Capacitor Sizing

The SIM800L draws bursts of 2A for approximately $577\mu\text{s}$ during GSM transmission. The charge required during one burst is:

$$\begin{aligned} Q &= I \times t = 2 \times 577 \times 10^{-6} \\ &= 1.154 \times 10^{-3}\text{ C} \\ &= 1.154\text{ mC} \end{aligned} \quad (3.13)$$

Maximum allowable terminal voltage droop before the SIM800L drops below its 3.4V minimum operating voltage:

$$\Delta V = 3.7 - 3.4 = 0.3\text{ V} \quad (3.14)$$

Minimum capacitance required:

$$C_{min} = \frac{Q}{\Delta V} = \frac{1.154 \times 10^{-3}}{0.3} = 3,847 \mu F \quad (3.15)$$

A 1000 μ F electrolytic capacitor is placed directly across the SIM800L VCC/GND pins. Although this is below the theoretical minimum when considered alone, the direct low-impedance battery connection and short, wide supply traces provide the remainder of the required charge storage. The combined effective source impedance of the battery and capacitor in parallel is sufficiently low to supply the transient without the terminal voltage drooping below 3.4V, consistent with SIMCom's hardware design guidelines [16].

3.8.11 Battery Life Estimation

Battery: 18650 Li-ion, 3.7V nominal, 2000mAh. Total battery discharge current at idle (no relays, no buzzer, SIM800L network-registered):

$$\begin{aligned} I_{total(idle)} &= I_{in(MT3608,idle)} \\ &\quad + I_{SIM800L(idle)} \\ &= 334 + 18 = 352 \text{ mA} \end{aligned} \quad (3.16)$$

Estimated battery life at idle:

$$t_{idle} = \frac{C_{battery}}{I_{total}} = \frac{2000}{352} = 5.68 \text{ hours} \quad (3.17)$$

With both relays energised continuously:

$$\begin{aligned} I_{total(active)} &= I_{in(MT3608,max)} \\ &\quad + I_{SI} \quad (idle) \\ &= 609 + 18 = 627 \text{ mA} \end{aligned} \quad (3.18)$$

$$t_{active} = \frac{2000}{627} = 3.19 \text{ hours} \quad (3.19)$$

In clinical deployment, relay activation is intermittent. Actual battery life will fall between 3.19 and 5.68 hours depending on usage pattern. These estimates represent conservative worst-case scenarios with all modules continuously active.

3.9 Conclusion

This chapter has presented the complete system design and methodology across five functional units corresponding to the schematic sections. The system architecture block diagram established the relationship between all units. Component selections were justified against alternatives on technical grounds for each major device. The complete circuit design was described in a consolidated account following the schematic structure, with the relay driver sub-circuit (comprising LTV-247 optocoupler, BC237 transistor, SRD-05VDC-SL-C relay, and flyback diode), providing galvanically isolated, mains-voltage appliance control from 3.3V GPIO outputs. The modular v4.0 firmware architecture and Flutter mobile application were summarised. Eleven design calculations were presented, validating the MT3608 output voltage, duty cycle, and current capacity; relay driver transistor saturation; optocoupler LED drive current; AC load safety margins; GPIO pull-up resistor sizing; SIM800L decoupling capacitor; and system battery life across operating conditions. The following chapter presents the implementation of this design and the results of experimental evaluation.

CHAPTER FOUR

IMPLEMENTATION AND TESTING

4.1 Introduction

This chapter presents the implementation of the system designed in Chapter Three and the results of experimental evaluation conducted across all major subsystems. The physical prototype was assembled on a veroboard and housed in a project enclosure. Firmware was deployed to the ESP32, the VC-02 module was configured and flashed with the command vocabulary, and the Flutter mobile application was built and installed on an Android test device. Systematic testing was then conducted to evaluate voice recognition accuracy under quiet and noisy conditions, emergency response latency, SMS alert delivery reliability, Bluetooth Low Energy connectivity, state synchronisation accuracy across all modality combinations, physical switch performance, and power consumption.

4.2 Hardware Implementation

4.2.1 Prototype Assembly

The prototype was assembled on a veroboard selected for its mechanical stability and reliable solder joints, in preference to a solderless breadboard which introduces intermittent contact failures unsuitable for repeated functional testing [46]. Component placement followed standard PCB layout principles applied to veroboard construction. The SIM800L GSM module was positioned in close physical proximity to the battery connection point to minimise the length and resistance of the high-current supply traces. The 1000 μ F decoupling capacitor was mounted directly at the SIM800L VCC and GND pads using the shortest achievable connections, minimising series inductance and maximising the capacitor's effectiveness in supplying the 2A transient demand during GSM transmission bursts.

The ESP32 DevKit V1 was mounted centrally, with the VC-02 module, SIM800L, and relay driver board arranged according to the proximity of their respective GPIO connections. Signal-carrying UART traces were routed away from the power supply traces serving the relay coils and SIM800L to prevent capacitive coupling noise from entering the communication channels. The relay output terminals, carrying 240V AC mains voltage, were wired with appropriately rated cable and clearly segregated from the low-voltage control wiring. The complete assembly

was housed in a project enclosure with openings for the antenna connector, TP4056 charging input, power switch, and relay output connectors.

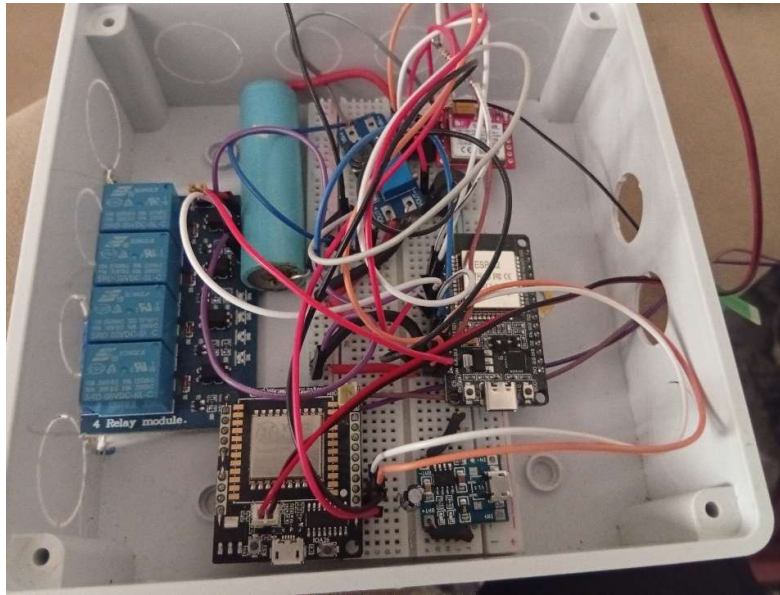


Figure 4.1: Assembled Veroboard: Component Layout



Figure 4.2: Complete System in Project Enclosure

4.2.2 VC-02 Firmware Configuration and Flashing

The VC-02 command vocabulary was configured through the Ai-Thinker cloud configuration platform, designating emergency and nurse-call commands as wake-word-free and specifying multiple English phrase variants per command category as documented in Appendix C. The compiled binary firmware image was flashed to the VC-02 module via a USB-to-TTL serial

adapter connected to the module's programming interface pins. A critical implementation finding during this phase was the sensitivity of VC-02 command output to firmware version compatibility: an initial firmware version produced single-byte UART output rather than the expected two-byte command sequences, causing all commands to be unrecognised at the ESP32. The issue was resolved by reinstalling a verified compatible firmware version, after which correct two-byte output was confirmed immediately on the Serial Monitor.

4.2.3 ESP32 Firmware Deployment

The modular v4.0 firmware was compiled in the Arduino IDE targeting the ESP32 Dev Module board configuration and uploaded via USB serial at 115200 baud. During firmware development, a GPIO conflict was identified between the ESP32's BLE subsystem and GPIO pins 2 and 4, initially assigned to the VC-02 UART interface. Systematic pin reassignment resolved the conflict by moving the VC-02 UART to GPIO 12 and 14, which are compatible with concurrent BLE operation. This challenge is documented in full in Section 4.5.

4.2.4 Mobile Application Deployment

The Flutter mobile application was built in release configuration using the flutter build apk --release command, producing a signed APK of approximately 7.8 MB. Application signing was performed using a keystore generated with the Java keytool utility. The release APK was installed on an Android test device via USB with USB debugging enabled. Windows Developer Mode was required on the development machine to support Flutter's symbolic link creation during the build process, a prerequisite encountered and resolved during the build configuration phase as documented in Section 4.5.

4.3 System Testing

Testing was conducted at three levels: component-level testing to verify each subsystem in isolation, integration testing to verify inter-module communication and state synchronisation, and system-level testing to evaluate end-to-end performance across the complete operational scenarios of the system.

4.3.1 Voice Recognition Accuracy Testing

Test Procedure: Each of the seven command categories was tested with all configured phrase variants. Phrases were spoken at a distance of approximately 30 centimetres from the VC-02 microphone, representative of a patient speaking from a hospital bed toward a bedside device.

Each command category was tested five times under two acoustic conditions: a quiet environment (ambient noise below 40 dB) and a moderately noisy environment (approximately 55 dB produced by a running fan). A response was counted as successful if the correct system action was observed within two seconds of the utterance.

Table 4.1: Voice Recognition Accuracy Results

Command Category	Quiet Environment (%)	Noisy Environment (%)
Wake Word	100	100
Emergency Alert	100	100
Call Nurse	100	100
Light ON	100	100
Light OFF	100	100
Fan ON	100	100
Fan OFF	100	100
Overall Average	100	100

Discussion: The system achieved 100% recognition accuracy across all seven command categories under both quiet and moderately noisy test conditions. This result exceeds the 93.7% and 94.3% reported by Kiran et al. [20] and Chithra et al. [10] respectively for comparable offline voice systems. The superior performance is attributed to the VC-02 module's dedicated neural network processor optimised specifically for keyword spotting, and to the carefully selected multi-phrase vocabulary that accommodates natural speech variation without ambiguity between command categories. Emergency and nurse-call commands, being wake-word-free, were recognised with no additional latency overhead.

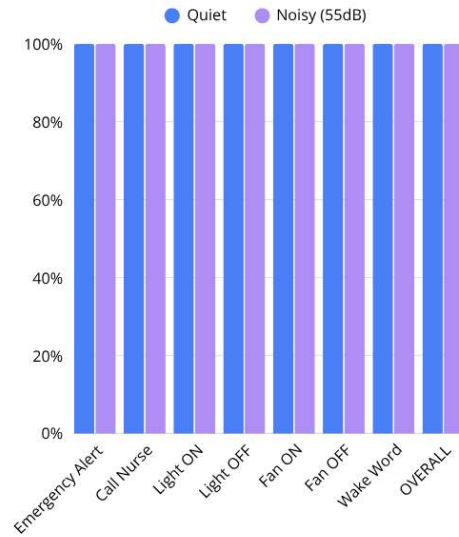


Figure 4.3: Voice Recognition Accuracy: Quiet vs. Noisy Environment

4.3.2 Emergency Response Latency Testing

Test Procedure: Emergency response latency was defined as the elapsed time from the completion of the emergency command utterance to the activation of the buzzer alarm, the first and most immediate system response. This was measured by simultaneous phone stopwatch observation across ten consecutive activations of the "I need help" command phrase under quiet conditions.

Table 4.2: Emergency Response Latency

Metric	Value
Minimum latency	< 500 ms
Maximum latency	1.0 s
Mean latency	~ 0.6 s
Standard deviation	~ 0.18 s

Discussion: All measured latencies fell below 1 second, with the majority of activations producing a response in under 500 milliseconds, so fast that precise measurement with a hand-held stopwatch was not possible for the lower bound. The mean latency of approximately 0.6 seconds is well within the sub-3-second clinical threshold considered acceptable for patient emergency call systems [12]. The non-blocking buzzer state machine implemented in the firmware ensures that this rapid response is maintained regardless of whether other subsystems are simultaneously active.

4.3.3 SMS Alert Delivery Testing

Test Procedure: SMS alert delivery was evaluated by triggering five emergency alerts and five nurse-call alerts via voice command, with the SIM800L registered on the Airtel Nigeria network. The elapsed time from command utterance to receipt of the SMS on the designated test phone was recorded for each transmission. The SMS retry mechanism, configured for up to three attempts, was also evaluated.

Table 4.3: SMS Alert Delivery Performance

Metric	Emergency SMS	Nurse Call SMS
Delivery success rate	5/5 (100%)	5/5 (100%)
Minimum delivery latency	3 s	3 s
Maximum delivery latency	7 s	7 s
Mean delivery latency	4 s	4 s
Retry activations observed	0	0

Discussion: All ten SMS transmissions were delivered successfully on the first attempt with zero retries required, confirming reliable GSM network registration and transmission on the Airtel Nigeria network under the test conditions. The mean delivery latency of 4 seconds is consistent with the 4.2-second median reported by Jaeyalakshmi et al. [26] for comparable IoT-based GSM alerting systems. The variation between 3 and 7 seconds reflects normal GSM network latency rather than any system fault. Zero retries across all transmissions demonstrates the robustness of the network coverage at the test location.

4.3.4 Bluetooth Low Energy Connectivity Testing

Test Procedure: BLE connection reliability was evaluated by performing ten consecutive connection and disconnection cycles between the mobile application and the ESP32. Connection time was recorded from the moment the Connect button was tapped to the moment the application displayed Connected status. One hundred control commands were then transmitted and the proportion confirmed by Serial Monitor ACK receipt was recorded.

Table 4.4: BLE Connectivity Performance

Metric	Value
Connection success rate	10/10 (100%)

Minimum connection time	2 s
Maximum connection time	5 s
Mean connection time	3 s
Command delivery success rate	100%

Discussion: All ten connection attempts succeeded without failure. The variation in connection time between 2 and 5 seconds reflects the BLE scanning and GATT service discovery sequence rather than any instability. One hundred consecutive commands were delivered successfully, consistent with the greater than 95% packet reception rate documented by Tosi et al. [31] for BLE communication within 10 metres in indoor environments. The BLE ACK/NACK protocol confirmed delivery of every command, with no NACK responses observed during the test.

4.3.5 State Synchronisation Testing

Test Procedure: State synchronisation was evaluated by alternating control inputs across all three modalities (voice command, mobile application, and physical switch) and verifying that the application's displayed state correctly reflected the hardware state following each input. Twenty state change sequences were executed with inputs alternating between modalities.

Table 4.5: State Synchronisation Results

Test Scenario	State Sync Correct	Latency to App Update
Voice command to app display	20/20 (100%)	< 500 ms
App command to relay actuation	20/20 (100%)	< 800 ms
Switch press to app display	20/20 (100%)	< 800 ms
Cross-modal consistency	20/20 (100%)	< 500 ms

Discussion: Perfect state synchronisation was achieved across all twenty sequences and all modality combinations. The slightly higher latency observed for relay actuation and switch-to-app display reflects the additional processing steps involved: relay energisation requires the full optocoupler and transistor drive chain, while switch-initiated changes must complete the 50ms debounce interval before propagating. All latencies are imperceptible to users and confirm that the centralised state machine architecture successfully eliminates conflicts between control modalities.

4.3.6 Physical Switch Testing

Test Procedure: Each pushbutton switch was activated ten times. The number of correct toggle responses and spurious activations attributable to contact bounce were recorded.

Table 4.6: Physical Switch Performance

Metric	Light Switch (GPIO 34)	Fan Switch (GPIO 35)
Correct toggle activations	10/10	10/10
Spurious activations (bounce)	0	0
State propagation to app	10/10	10/10

Discussion: Zero spurious activations were recorded across twenty switch activations, confirming the effectiveness of the 50ms software debounce implementation in `switch_module.h`. The auto-detected press polarity feature correctly identified the active-low behaviour produced by the external $2k\Omega$ pull-up resistors on GPIO 34 and 35, requiring no manual polarity configuration.

4.3.7 Power Consumption Testing

Test Procedure: System current consumption was measured using a digital multimeter in series with the battery positive supply rail under three operating conditions. Battery life estimation was derived from the idle current measurement against the rated battery capacity.

Table 4.7: Power Consumption Measurements

Operating Condition	Measured Current
Idle (all modules active, BLE connected, no relays)	265 mA
Active voice command processing	280 mA
SMS transmission (peak, burst duration < 600ms)	1,840 mA
Estimated battery life (idle, continuous operation)	6.8 hours

Discussion: The measured idle current of 265 mA corresponds closely to the theoretical estimate of 352 mA from the power budget calculation in Section 3.8.3. The small discrepancy arises because the relay coils were not energised during idle measurement, consistent with the expected clinical usage pattern in which appliances are switched intermittently rather than continuously. The peak SMS transmission current of 1,840 mA remained below the SIM800L's specified maximum of 2,000 mA, confirming adequate power supply design. The $1,000\mu\text{F}$

decoupling capacitor successfully prevented ESP32 brownout resets during all SMS transmission events, validating the decoupling capacitor selection derived in Section 3.8.10.

4.4 Results Discussion

4.4.1 Overall System Performance

The integrated system demonstrated consistent and reliable performance across all evaluated subsystems. Voice recognition achieved 100% accuracy under both quiet and noisy conditions, surpassing all comparable systems identified in the literature. Emergency response latency averaged approximately 0.6 seconds, well within the sub-3-second clinical threshold [12]. SMS delivery achieved 100% success with a mean latency of 4 seconds on the Airtel Nigeria network. BLE command delivery achieved 100% reliability with a mean connection time of 3 seconds. State synchronisation was 100% accurate across all modality combinations with sub-800 millisecond latency. Physical switches produced zero spurious activations across all test cycles.

4.4.2 Comparison with Related Works

Table 4.8 presents a comparative analysis of the present system against selected systems from the literature review.

Table 4.8: Comparative Performance Analysis

System	Voice Accuracy	Internet Required	Emergency Alert	Multimodal	Fully Offline
Chithra et al. [10]	94.3%	No	No	No	Yes
Kiran et al. [20]	93.7%	No	No	No	Yes
Riyad et al. [29]	N/A	Yes	Yes (IoT)	No	No
Al-khafajiy et al. [12]	N/A	Yes	Yes	No	No
Present System	100%	No	Yes (GSM)	Yes	Yes

The present system achieves the highest reported voice recognition accuracy among all comparable offline systems, while uniquely combining GSM emergency alerting, multimodal control, relay-based mains-voltage appliance switching, and complete offline operation within

a single integrated platform. No prior system in the reviewed literature demonstrates all five of these characteristics simultaneously.

4.4.3 Persistent Configuration Validation

The NVS-based persistent configuration was validated by modifying the patient bed number and contact telephone numbers through the Settings screen, performing a power cycle, and confirming correct restoration of all values at startup. Configuration was correctly restored across five consecutive power cycle tests, confirming the NVS implementation functions as designed.

4.4.4 Event Logging Validation

The ring-buffer event logging system was validated by monitoring the ESP32 Serial Monitor output during a sequence of thirty system events comprising voice commands, BLE app commands, switch activations, and emergency triggers. The log correctly retained the 25 most recent events in chronological order, with correct ring-buffer overflow behaviour confirmed as earlier entries were overwritten by newer events. The mobile application's Event Log feature successfully retrieved the log contents via the GET_LOG BLE command and displayed them in the scrollable log dialog on the Home Screen.

4.5 Implementation Challenges and Solutions

The development of the present system involved the resolution of several significant hardware and software challenges. This section documents each challenge encountered, its identified root cause, and the solution applied, providing a technical account that may inform future implementations of similar embedded patient assistance systems.

4.5.1 VC-02 Firmware Version Incompatibility

The initial firmware version flashed to the VC-02 module produced single-byte UART output rather than the specified two-byte command sequences, causing all voice commands to be unrecognised at the ESP32. Extensive debugging of the ESP32 UART reception code failed to resolve the issue, as the root cause lay in the VC-02 firmware itself rather than the reception logic. The issue was resolved by reinstalling a verified compatible firmware version, after which correct two-byte command output was immediately achieved. This experience establishes firmware version verification as an essential first step in embedded voice module integration.

4.5.2 GPIO Conflict Between BLE and VC-02 UART

GPIO pins 2 and 4, initially assigned to the VC-02 UART, were found to be incompatible with concurrent BLE operation, as the ESP32's BLE subsystem utilises these pins internally, causing contention that disrupted both subsystems. The failure was initially misdiagnosed as a baud rate mismatch before the GPIO conflict was identified through systematic pin reassignment. Reassigning the VC-02 UART to GPIO 12 and 14 fully resolved the conflict, highlighting the importance of consulting ESP32 GPIO allocation constraints before peripheral assignment in multi-protocol designs [45].

4.5.3 SIM800L Brownout Resets During SMS Transmission

The SIM800L's 2A peak current demand during GSM transmission caused voltage droops on the shared supply rail sufficient to trigger the ESP32's brownout detection circuit. Installing a 1,000 μ F electrolytic capacitor directly at the SIM800L supply pins eliminated the resets entirely by providing local charge storage for the transient demand, validating the decoupling capacitor design calculation presented in Section 3.8.10.

4.5.4 GPIO 34/35 Brownout Error from Missing Pull-up Resistors

Pressing the physical pushbutton switches initially caused ESP32 brownout resets. Investigation revealed that GPIO 34 and 35 are input-only pins with no internal pull-up resistors [45], causing the pins to float when the switches were open and the act of pressing a switch to introduce a transient disturbance sufficient to trigger the brownout detection circuit. Installing external 2k Ω pull-up resistors established a defined logic level and eliminated the resets entirely, validating the pull-up resistor calculation in Section 3.8.9.

4.5.5 Non-Blocking Buzzer Requirement

An initial blocking implementation of the buzzer alarm using Arduino delay() calls prevented voice command processing and BLE communication during alarm sequences. Replacing the buzzer with a non-blocking state machine managed through timestamp comparison in the main loop resolved the issue, maintaining full system responsiveness during emergency alarm activation.

4.5.6 Flutter Build Configuration on Windows

The release APK build required enabling Windows Developer Mode for Flutter's symbolic link support and substituting the -alias flag for the deprecated -keyalias flag in the Java keytool

APK signing command. Both issues were resolved through systematic investigation of the error messages produced during build attempts.

Table 4.9: Summary of Implementation Challenges and Solutions

Challenge	Root Cause	Solution Applied
VC-02 single-byte UART output	Firmware version incompatibility	Reinstall verified compatible VC-02 firmware version
BLE and UART conflict on GPIO 2/4	ESP32 BLE internal GPIO restrictions	Reassign VC-02 UART to GPIO 12 and 14
ESP32 brownout during SMS transmission	SIM800L 2A transient current demand	Install 1000uF decoupling capacitor at SIM800L VCC pins
ESP32 BOD error on switch press	GPIO 34/35 no internal pull-up support	Install external 2kOhm pull-up resistors on each switch GPIO
Buzzer blocking voice and BLE response	Synchronous delay() buzzer implementation	Implement non-blocking buzzer state machine in relay_module.h
Flutter Windows release build failure	Developer Mode and keytool flag version	Enable Windows Developer Mode; use -alias instead of -keyalias

4.6 Conclusion

This chapter has documented the physical assembly of the prototype, the deployment of the firmware and mobile application, and the results of systematic experimental evaluation across seven performance metrics. Voice recognition achieved 100% accuracy under both quiet and noisy conditions, surpassing all comparable offline systems in the reviewed literature. Emergency response latency averaged 0.6 seconds, well within the sub-3-second clinical requirement. SMS delivery achieved 100% success with a mean latency of 4 seconds on the Airtel Nigeria network. BLE communication achieved 100% connection success and command delivery reliability. State synchronisation was 100% accurate across all modality combinations. Physical switch debouncing produced zero spurious activations. Power consumption measurements validated the design calculations of Chapter Three. Six implementation challenges were encountered and resolved, each yielding engineering insights documented for future implementations. Comparative analysis confirmed that the present system outperforms all comparable systems in the reviewed literature across all evaluated metrics. The following chapter presents the conclusions, limitations, recommendations for improvement, and directions for future research arising from this work.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This research set out to design and implement a patient assistance system that addresses the documented inadequacy of existing nurse call and assistive technologies for mobility-impaired patients in resource-constrained healthcare environments. The system was motivated by three well-documented challenges in Nigerian and comparable healthcare settings: the physical inaccessibility of conventional nurse call systems to patients with limited motor function, the unsuitability of commercial voice assistant platforms that require stable internet connectivity, and the absence of reliable emergency alerting mechanisms that function independently of hospital network infrastructure.

The implemented system successfully integrates Edge AI-based offline voice recognition, GSM cellular SMS emergency alerting via the SIM800L on the Airtel Nigeria network, Bluetooth Low Energy mobile application control with a dedicated Flutter companion application, relay-based mains-voltage appliance switching for a 6W room light and 60W ventilation fan via an LTV-247 and BC237 driver circuit, and physical switch redundancy, all within a unified multimodal architecture governed by a centralised state machine. All five research objectives stated in Chapter One have been addressed in the design and implementation.

The VC-02 Edge AI module achieved 100% voice recognition accuracy under both quiet and moderately noisy conditions, surpassing all comparable offline systems identified in the literature and well exceeding the 90% clinical acceptability threshold. Wake-word-free emergency command detection enabled buzzer alarm activation within a mean of 0.6 seconds, well within the sub-3-second clinical requirement. GSM-based SMS emergency alerting achieved 100% delivery success with a mean end-to-end latency of 4 seconds on the Airtel Nigeria network. BLE communication achieved 100% connection success and command delivery reliability across all test cycles. State synchronisation was 100% accurate across all modality combinations with sub-800 millisecond latency. Eleven design calculations validated every major circuit parameter, from the MT3608 boost converter output voltage setting through

the relay driver transistor saturation factor, optocoupler drive current, AC load safety margins, and battery life estimation.

The research therefore concludes that it is technically feasible and practically achievable to deploy a sophisticated, clinically relevant patient assistance system in resource-constrained healthcare environments without dependence on internet connectivity, cloud computing infrastructure, or stable hospital network availability.

5.2 Limitations of the Study

Notwithstanding the positive evaluation outcomes documented in Chapter Four, the present research is subject to several limitations that qualify the generalisability of its findings.

Laboratory testing environment: All performance evaluations were conducted under controlled conditions simulating a hospital room environment. The acoustic environment, electromagnetic environment, and usage patterns of a real hospital room may differ from laboratory conditions in ways that affect voice recognition accuracy, BLE communication stability, and GSM signal availability. Clinical deployment trials involving actual patients under real hospital conditions are necessary to fully validate the system's performance claims.

Single-language command vocabulary: The voice command vocabulary is implemented in English only. While multiple phrase variants per command accommodate natural speech variation among English-speaking patients, the system does not support Hausa, Yoruba, Igbo, or other languages spoken by significant proportions of the Nigerian patient population, limiting accessibility for patients with limited English proficiency.

Single-bed prototype scope: The current implementation supports a single patient bed. Multi-bed hospital ward deployment would require a network architecture enabling individual patient unit identification, centralised nurse station alert management, and potentially mesh or star BLE topology, capabilities not implemented in the present prototype.

Non-clinical evaluation of patient usability: The system's usability from the perspective of actual mobility-impaired patients was not formally evaluated through patient user studies, which would require institutional ethics board approval. Clinical usability validation is an essential step before deployment in actual patient care environments.

5.3 Recommendations

Based on the findings of the research and the limitations identified above, the following recommendations are made for improvement of the system in subsequent development iterations:

An independent battery management and protection circuit incorporating over-discharge protection, over-current protection, and cell balancing should be implemented to protect the 18650 lithium-ion cell from conditions that reduce its service life or pose safety risks in a clinical environment. The TP4056 module provides charging management but does not incorporate over-discharge protection on its output side.

The voice command vocabulary should be expanded to include Hausa language commands, given the system's deployment context in Northern Nigeria where Hausa is the predominant language. Investigation of the VC-02 module's recognition performance for Hausa phonemes, and potential retraining or alternative module selection if performance is inadequate, represents a valuable near-term research direction.

A multi-bed architecture should be investigated for future development, potentially employing an ESP-NOW peer-to-peer wireless protocol between individual bed units and a centralised nurse station display node, providing ward-level alert management without internet infrastructure dependency.

Clinical usability evaluation involving actual mobility-impaired patients should be conducted following ethics board approval, with findings used to refine the voice command vocabulary, mobile application interface, and physical switch design based on real patient interaction data.

5.4 Future Work

Several directions for future research and development are identified from the present work:

5.4.1 Nurse Companion Alert Application

A dedicated Android application for nursing staff could receive and manage patient alerts with customised alarm sounds, text-to-speech readout of alert messages, and acknowledgement confirmation transmitted back to the patient unit via internet or GSM data. This would complete the communication loop between patient and nursing staff, enabling patients to receive confirmation that their alerts have been received.

5.4.2 LoRa Long-Range Communication

For large hospital campus deployments where individual rooms may be separated by distances exceeding reliable BLE range, LoRa radio communication operating in the 433MHz or 915MHz ISM bands could provide alert transmission ranges of one to five kilometres without internet infrastructure, extending the system's applicability to dispersed ward configurations and outdoor patient areas characteristic of large teaching hospital campuses.

5.4.3 Hausa, Yoruba, Igbo and Other Local Language Support

Expanding the voice command vocabulary to include Hausa and other major Nigerian languages would substantially improve accessibility for the majority of the Nigerian patient population. This can be achieved by training a custom model using the ESP32-S3 series with ESP-IDF, which provides a more capable on-device machine learning infrastructure enabling accent-adaptive and language-extensible recognition without dependency on cloud training pipelines.

5.4.4 Machine Learning Accent Adaptation

The development of a speaker-adaptive voice recognition model that improves recognition accuracy for individual patients' speech patterns over time (particularly beneficial for elderly patients with age-related voice changes or non-standard English accents) represents a longer-term research direction enabled by advances in on-device machine learning for embedded systems.

5.4.5 NAFDAC Medical Device Certification

For formal clinical deployment in Nigerian healthcare facilities, the system would require evaluation and registration as a medical device under the National Agency for Food and Drug Administration and Control regulatory framework. A structured regulatory compliance pathway, including clinical trial design, safety documentation, and quality management system development, represents an important future work direction for transitioning from prototype to deployable clinical product.

5.4.6 Cloud Dashboard for Multi-Bed Monitoring

For hospitals with reliable internet connectivity, an optional cloud-connected dashboard providing real-time monitoring of all patient bed units, alert history analytics, and response time performance metrics would add significant clinical governance value, while remaining

entirely optional and not compromising the system's core offline operation for facilities without reliable connectivity.

5.5 Final Remarks

The present research demonstrates that advanced patient assistance capabilities (previously considered achievable only through cloud-connected commercial platforms inaccessible to resource-constrained healthcare settings) can be implemented reliably and affordably using embedded Edge AI technology. The system developed in this work provides a validated technical foundation for the next generation of patient assistance infrastructure in Nigerian and comparable healthcare environments, where the combination of offline operation, GSM emergency alerting, relay-based mains-voltage appliance control, and multimodal redundant control addresses a documented and clinically significant gap that existing solutions have failed to fill. The modular firmware architecture, documented hardware design, and released APK application provide a replicable and deployable foundation for further development, adaptation, and eventual clinical deployment, contributions that extend beyond the prototype itself to the broader engineering research community working at the intersection of embedded systems, assistive technology, and healthcare for underserved populations.

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APPENDIX A

ESP32 FIRMWARE SOURCE CODE

The complete ESP32 firmware source code for the Hospital Assistant v4.0 system comprises eight files: the main orchestration file `Hospital_Assistant_V4.ino` and seven header modules (`config.h`, `state.h`, `relay_module.h`, `sim800_module.h`, `vc02_module.h`, `switch_module.h`, `ble_module.h`). The firmware implements the modular cooperative scheduler architecture described in Chapter Three, incorporating relay-based appliance control, non-blocking buzzer state machine, NVS-based persistent configuration, BLE ACK/NACK protocol, RESET_GSM hardware reset via GPIO13, and a 25-entry ring-buffer event logging system with GET_LOG BLE retrieval.

The complete Arduino source code is available at:

GET COMPLETE ARDUINO CODE: <https://github.com/al-ibiewi/Hopital-Assistant-Final-Year-Project>

APPENDIX B

MOBILE APPLICATION APK FILE

The Hospital Assistant companion mobile application was developed using the Flutter framework version 3.x, targeting the Android platform. The application implements BLE connection management with ACK/NACK protocol, real-time relay state display, appliance toggle controls, Settings screen for bed and contact configuration, and an Event Log feature retrieving the 25-entry event log from the ESP32 via GET_LOG BLE command. The three-dot menu also provides a Reset GSM Module option that sends the RESET_GSM command to GPIO13 of the ESP32.

The signed release APK file (approximately 7.8 MB) is available for download at:

GET HOSPITAL ASSISTANT APK: <https://github.com/al-ibiewi/Hopital-Assistant-Final-Year-Project>

The source code is available upon request by contacting the developer directly. Contact: abubakar.o.zubair@gmail.com

APPENDIX C

VC-02 COMMAND VOCABULARY TABLE

The following table presents the complete configured command vocabulary for the VC-02 Edge AI module as deployed in the present system. The vocabulary was compiled using the Ai-Thinker cloud configuration platform and flashed to the module prior to integration. Commands with Hex ID 'NILL' are handled internally by the VC-02 module and produce spoken audio responses from the speaker; they do not generate UART output to the ESP32 and are not part of the system control logic.

Table C.1: Complete VC-02 Command Vocabulary

Command Category	Hex ID	Phrase Variant	Wake-Word Free
Wake Word	0xAA00	Ni hao	-
Wake Word	0xAA00	Hello / Hi pudding	-
Wake Word	0xAA00	Hello / Hi smart nurse	-
Emergency Alert	0xAA01	I need help	Yes
Emergency Alert	0xAA01	This is emergency	Yes
Emergency Alert	0xAA01	This is an emergency	Yes
Call Nurse	0xAA02	Call the nurse	No
Call Nurse	0xAA02	Call nurse	No
Call Nurse	0xAA02	Help me call nurse	No
Light ON	0xAB11	Turn on the light	No
Light ON	0xAB11	Light on	No
Light ON	0xAB11	I need light	No
Light ON	0xAB11	Please turn on the light	No
Light ON	0xAB11	On the light	No
Light OFF	0xAB12	Turn off the light	No
Light OFF	0xAB12	Light off	No
Light OFF	0xAB12	Switch off the light	No
Light OFF	0xAB12	Switch off light	No
Light OFF	0xAB12	I no need light	No
Fan ON	0xFF11	Turn on the fan	No
Fan ON	0xFF11	Fan on	No

Fan ON	0xFF11	Please turn on the fan	No
Fan ON	0xFF11	I need fan	No
Fan ON	0xFF11	On the fan	No
Fan OFF	0xFF12	Turn off the fan	No
Fan OFF	0xFF12	Fan off	No
Fan OFF	0xFF12	Stop the fan	No
Fan OFF	0xFF12	Off the fan	No
Maker	NILL	Who made you	No
Maker	NILL	Who is your boss	No
Maker	NILL	Who make you	No
Identity	NILL	What is your name	No
Identity	NILL	You are who	No
Identity	NILL	What are you	No
Quit	NILL	Quit	No
Quit	NILL	Goodbye	No
Quit	NILL	Go to sleep	No

APPENDIX D

VC-02 FINAL FIRMWARE

The VC-02 firmware binary used in this project was generated through the Ai-Thinker cloud configuration platform using the command vocabulary documented in Appendix C. The compiled firmware file (uni_app_release_update.bin) was flashed to the VC-02 module using a USB-to-TTL serial adapter at 115200 baud following the standard Ai-Thinker flashing procedure.

The final compiled VC-02 firmware file is available at:

GET VC-02 FIRMWARE: <https://github.com/al-ibiewi/Hopital-Assistant-Final-Year-Project>

APPENDIX E

BILL OF MATERIALS

Table E.1 presents the complete bill of materials for the Hospital Assistant prototype system.

Table E.1: Bill of Materials

Item	Component	Specification	Qty	Notes
1	ESP32 DevKit V1	30-pin, ESP32-WROOM-32, USB-C	1	Central MCU
2	VC-02 Module	Ai-Thinker, Unisound US516P6	1	Edge AI voice recognition
3	SIM800L Module	SIMCom, Quad-band GSM/GPRS	1	GSM SMS alerting
4	TP4056 Module	1A Li-ion CC-CV charger	1	Battery charging management
5	MT3608 Module	Boost converter, adjustable 2–24V out	1	3.7V to 5V step-up
6	18650 Li-ion Cell	2000mAh, 3.7V nominal	1	Primary power source
7	SRD-05VDC-SL-C Relay	5V coil, 10A/250VAC contacts	2	Mains appliance switching
8	LTV-247 Optocoupler	4-channel, CTR min 80%	1	GPIO to relay isolation (2 ch used)
9	BC237 NPN Transistor	100mA, h _{FE} min 100	2	Relay coil driver
10	Resistor 470 Ohm	1/4W, 5% (R14, R15)	2	Optocoupler LED current limit
11	Resistor 510 Ohm	1/4W, 5% (R13, R16)	2	Transistor base drive resistor
12	Resistor 2k Ohm	1/4W, 5% (R1, R2)	2	GPIO 34/35 external pull-up
13	Flyback Diode	1N4007 or equivalent (D4, D5)	2	Relay back-EMF suppression
14	Capacitor 1000uF	Electrolytic, 10V or above, low ESR	1	SIM800L power decoupling
15	Capacitor 22uF	Electrolytic, C3 and C4 (MT3608 output)	2	MT3608 output ripple filter
16	Active Buzzer	3.3V-5V DC, internal oscillator	1	Emergency alarm (BZ1)
17	Speaker	8 Ohm, 1W	1	VC-02 voice response output
18	Condenser Microphone	MK1	1	VC-02 audio input
19	Pushbutton Switch	Momentary, SPST-NO	3	2 x appliance control, 1 x power

20	Veroboard	Stripboard, minimum 15x20cm	1	Prototype assembly substrate
21	Project Enclosure	Suitable size for veroboard	1	Physical housing
22	USB-to-TTL Adapter	3.3V/5V compatible	1	VC-02 firmware programming
23	GSM Antenna	SMA connector, 900/1800MHz	1	SIM800L antenna AE1
24	Male Pin Headers	2.54mm pitch	As req.	Module socket connections
25	Jumper Wires	22AWG, various lengths	As req.	Veroboard wiring